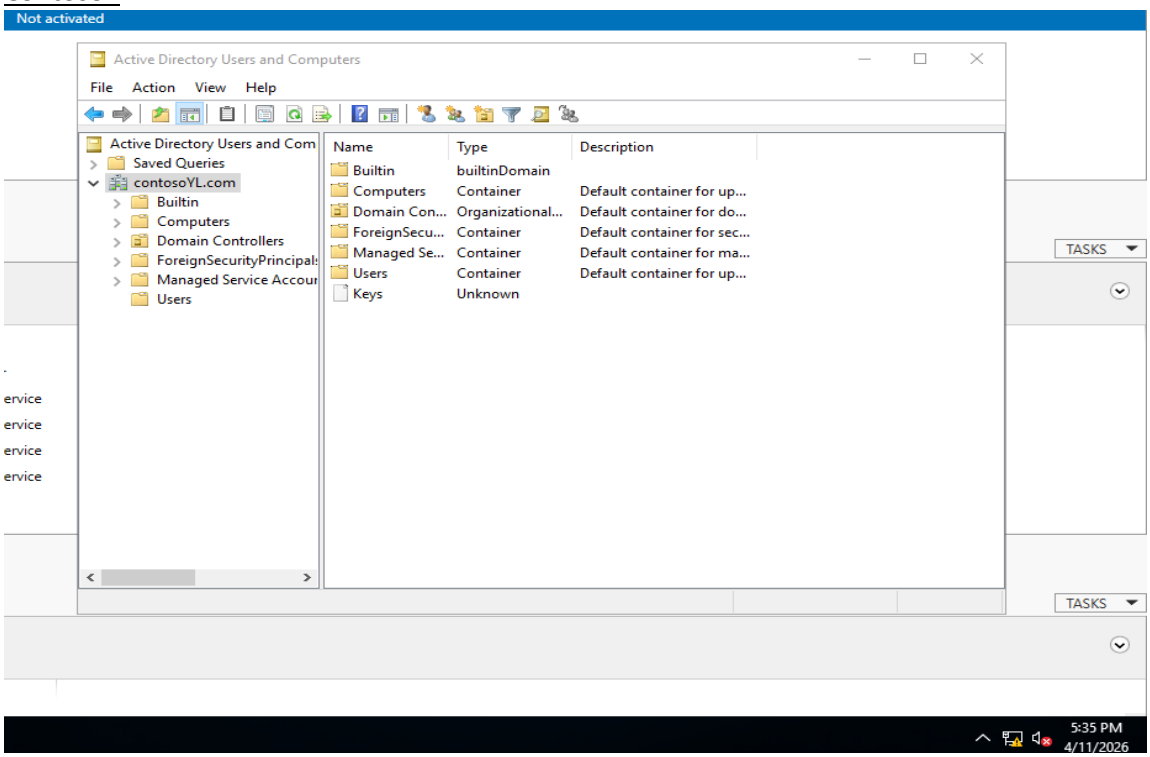
Travail Pratique **Youssef Lahlali**

SRV-WIN(III) **420-4RS-HY**

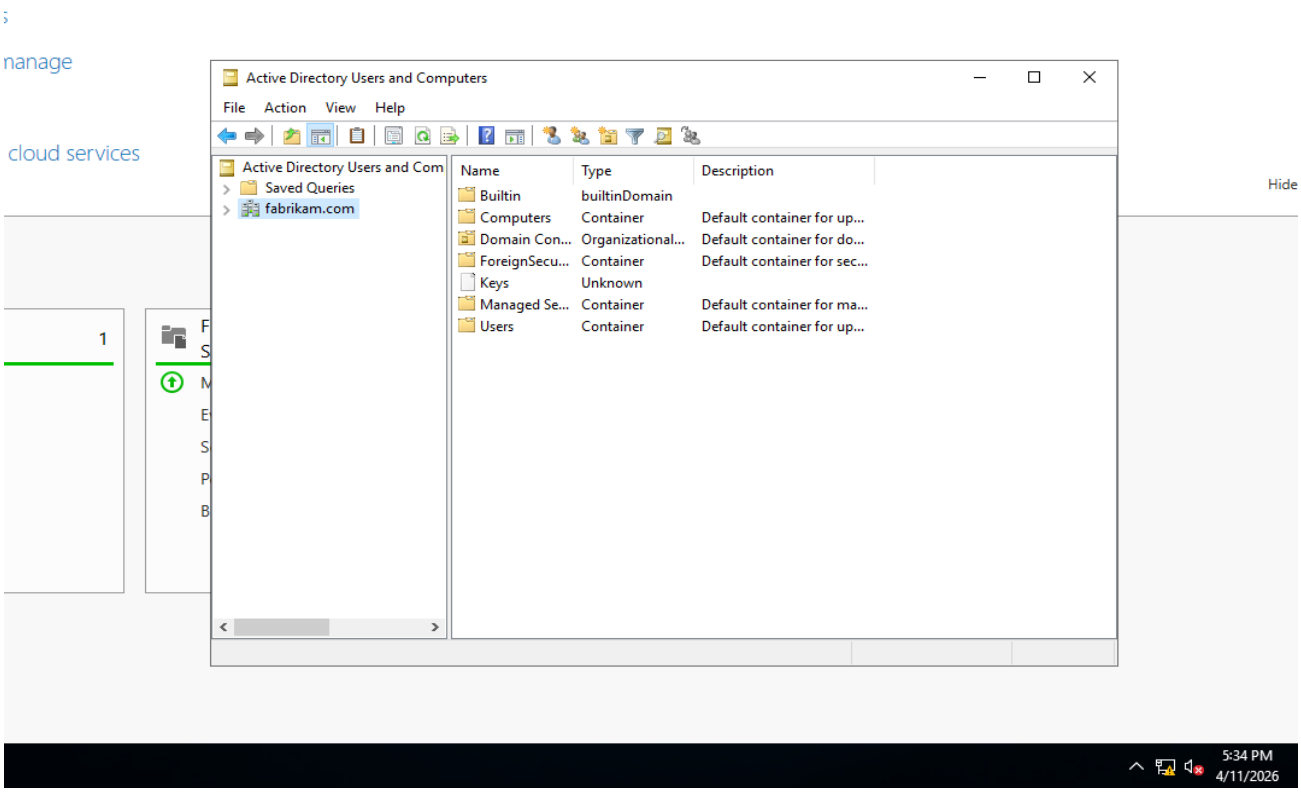
Section 1 – Préparation (10 pts)

1. Installez et configurez les deux domaines/forêts indépendants.

Contoso :

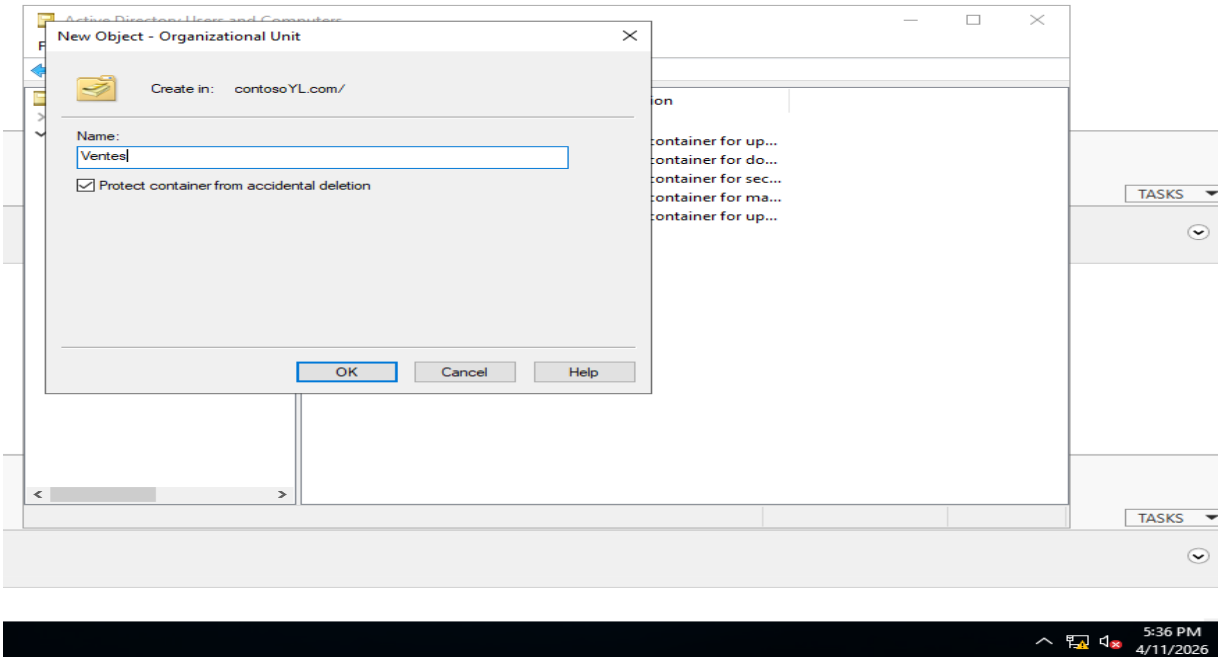


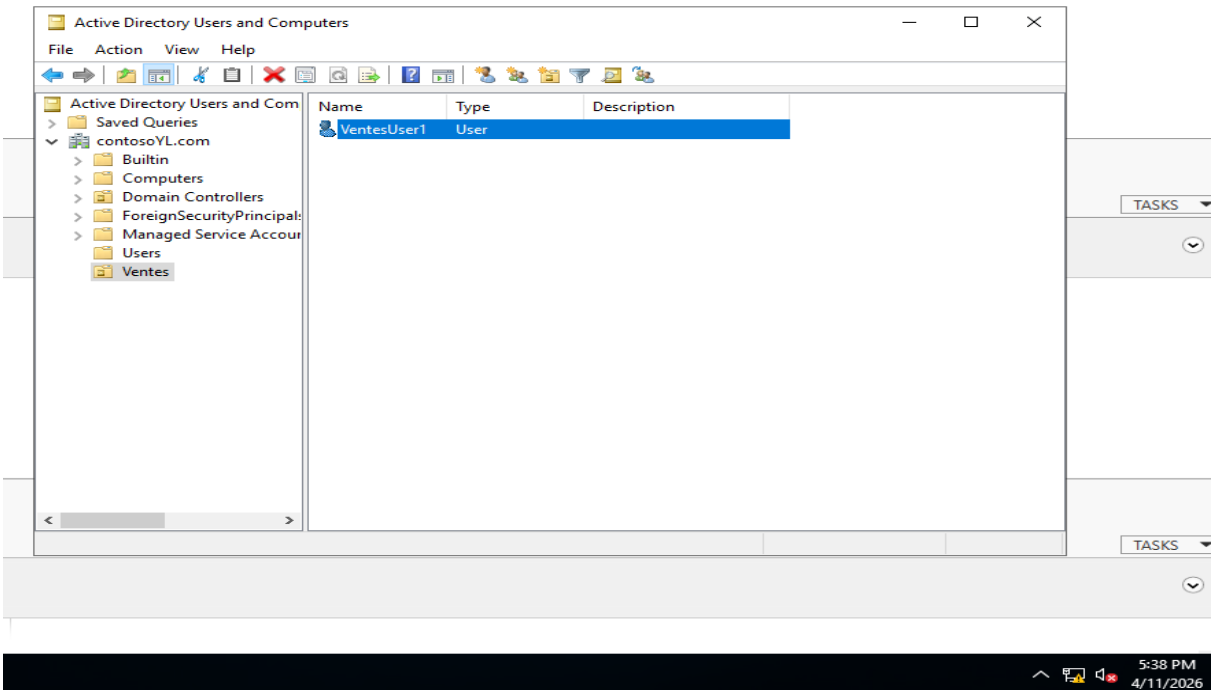
Fabrikam :


nanage
cloud services
Hide
Active Directory Users and Computers
File Action View Help
Active Directory Users and Com
Saved Queries
fabrikam.com
Name Type Description
Builtin builtinDomain
Computers Container Default container for up...
Domain Con... Organizational... Default container for do...
ForeignSecu... Container Default container for sec...
Keys Unknown
Managed Se... Container Default container for ma...
Users Container Default container for up...
5:34 PM
4/11/2026

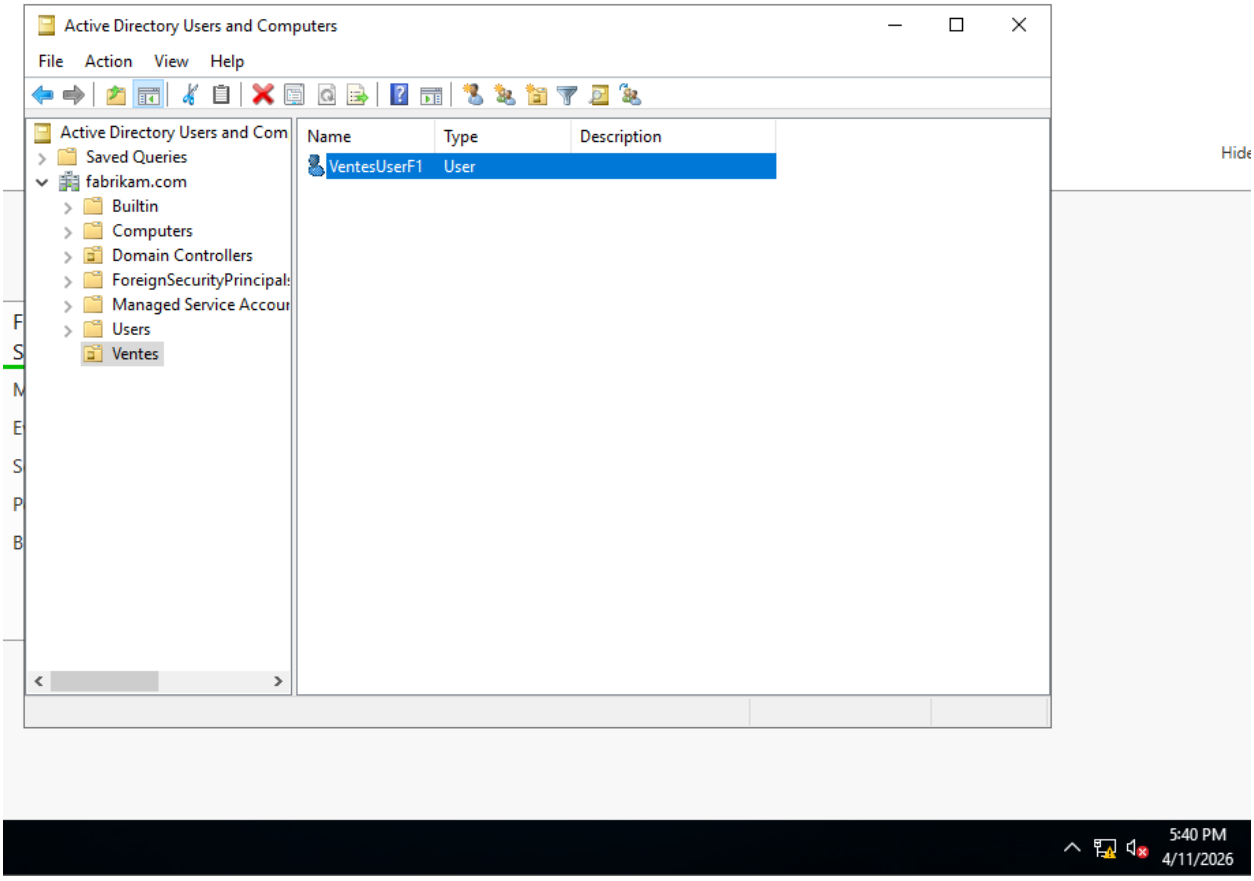
2. **Créez les OU et utilisateurs suivants : ○**

Contoso.com : OU = Ventes, utilisateur = VentesUser1





○ **Fabrikam.com : OU = Ventes, utilisateur = VentesUserF1**



3. Joignez un client au domaine fabrikam.com et un autre à Contoso.com et testez la connexion.

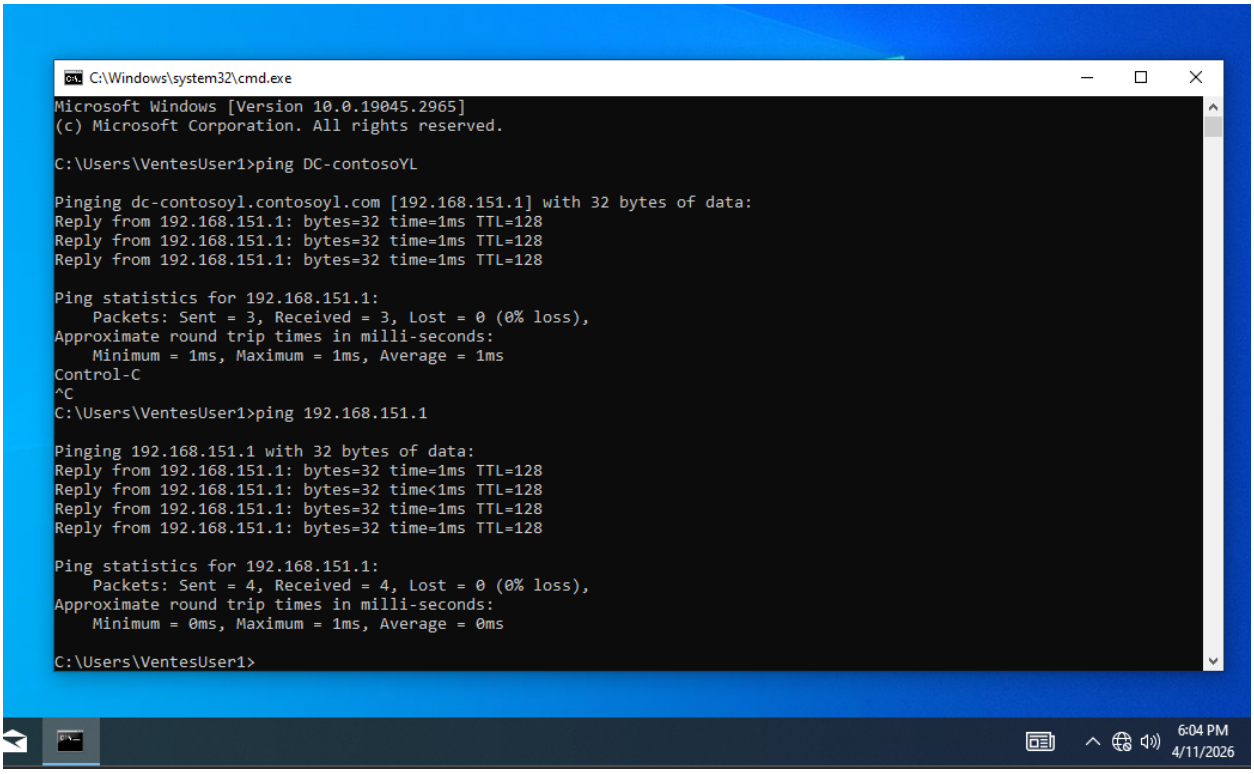
Ipconfig /all DC-Constoso :

```
   Host Name . . . . . . . . . . . . : DC-contosoYL
   Primary Dns Suffix  . . . . . . . : contosoYL.com
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : contosoYL.com

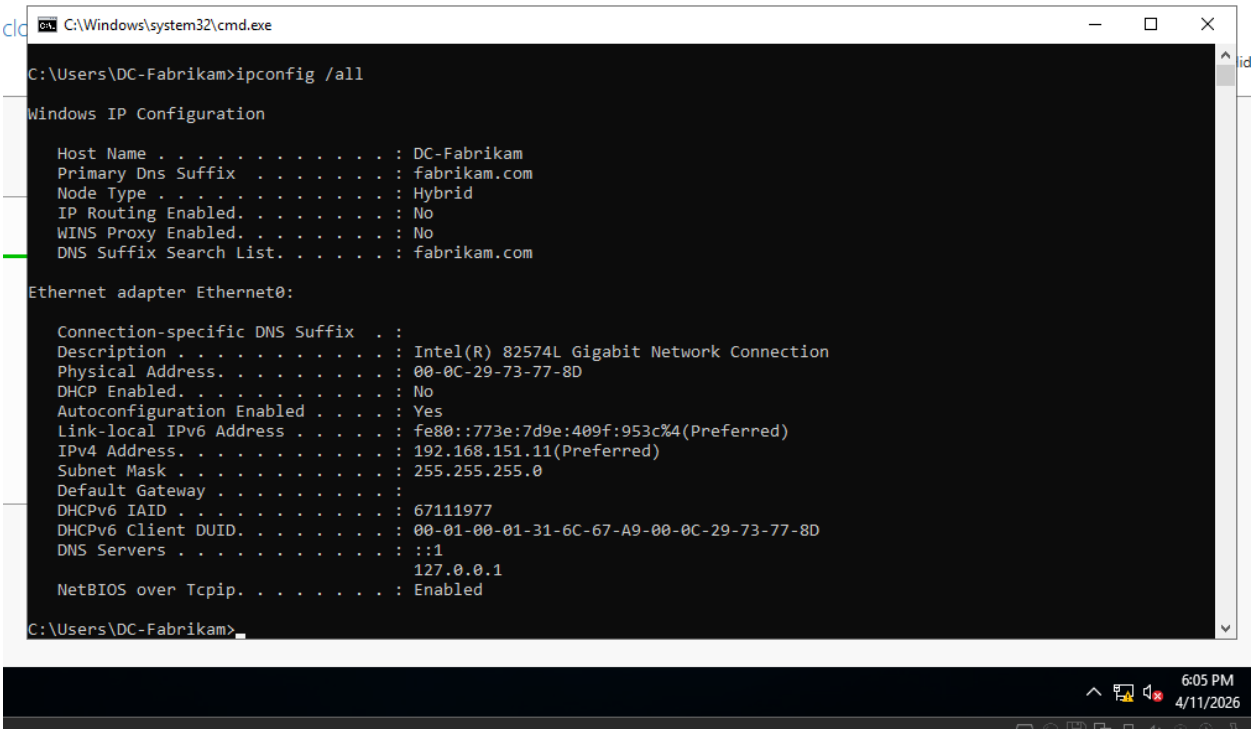
Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-0C-29-EF-16-1D
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::3bfa:25b6:a420:4007%4(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.151.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 67111977
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-31-02-A9-00-0C-29-EF-16-1D
   DNS Servers . . . . . . . . . . . : ::1
                                       127.0.0.1
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

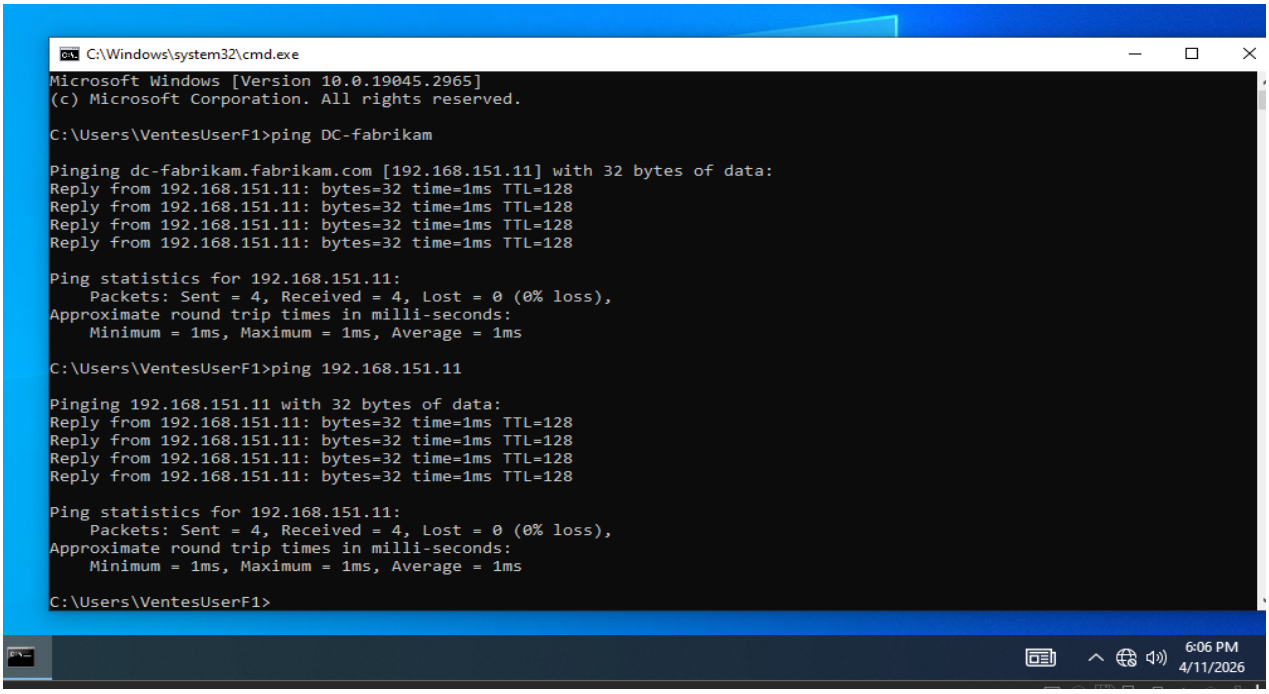
PING du VentesUser1 vers DC-Contoso



Ipconfig /all du DC-Fabrikam :

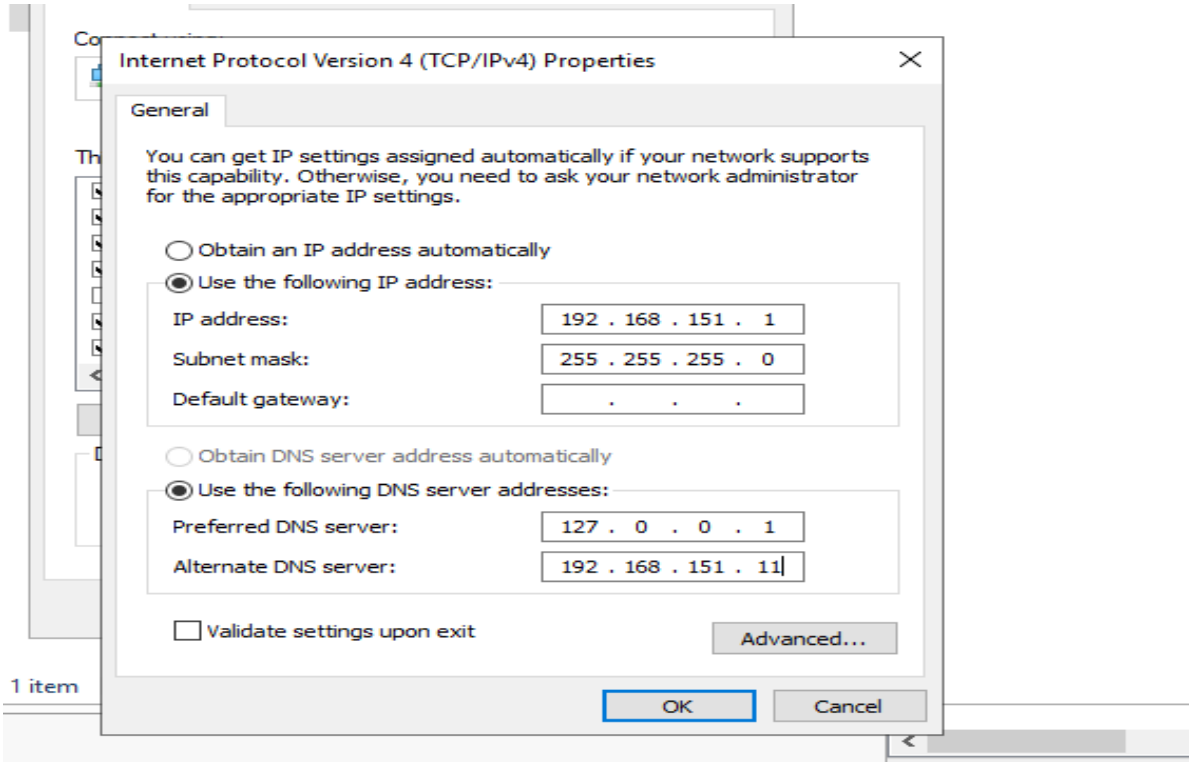


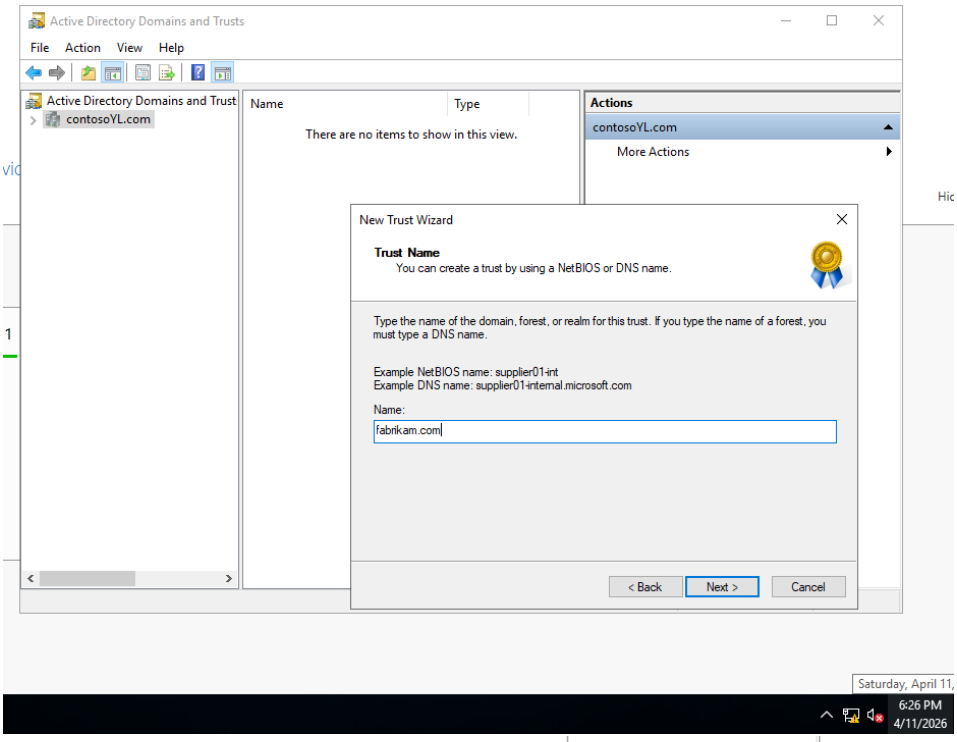
PING du VentesUserF vers DC-Contoso :

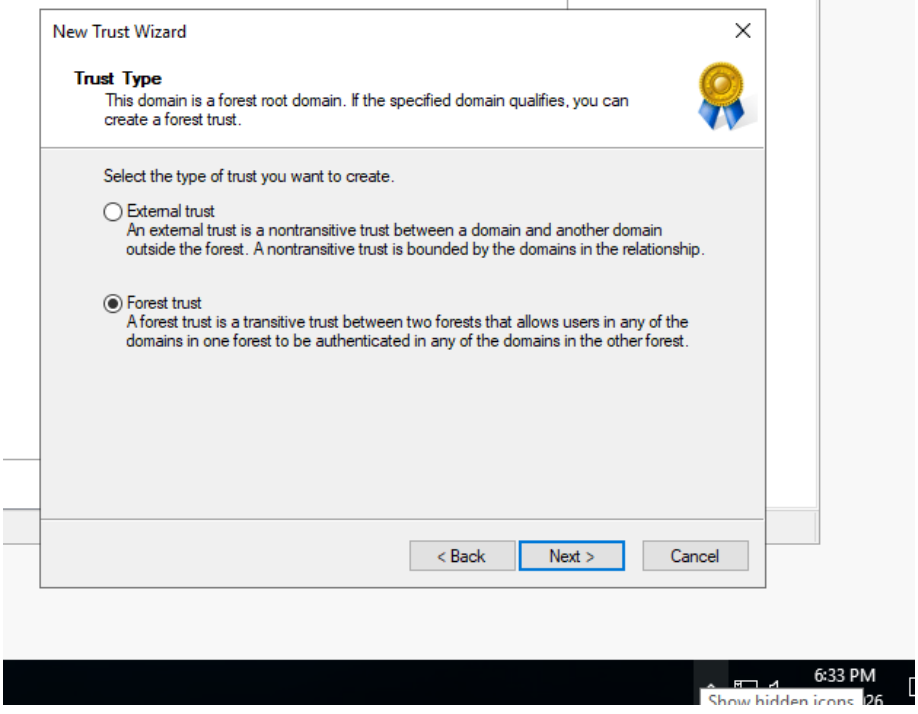


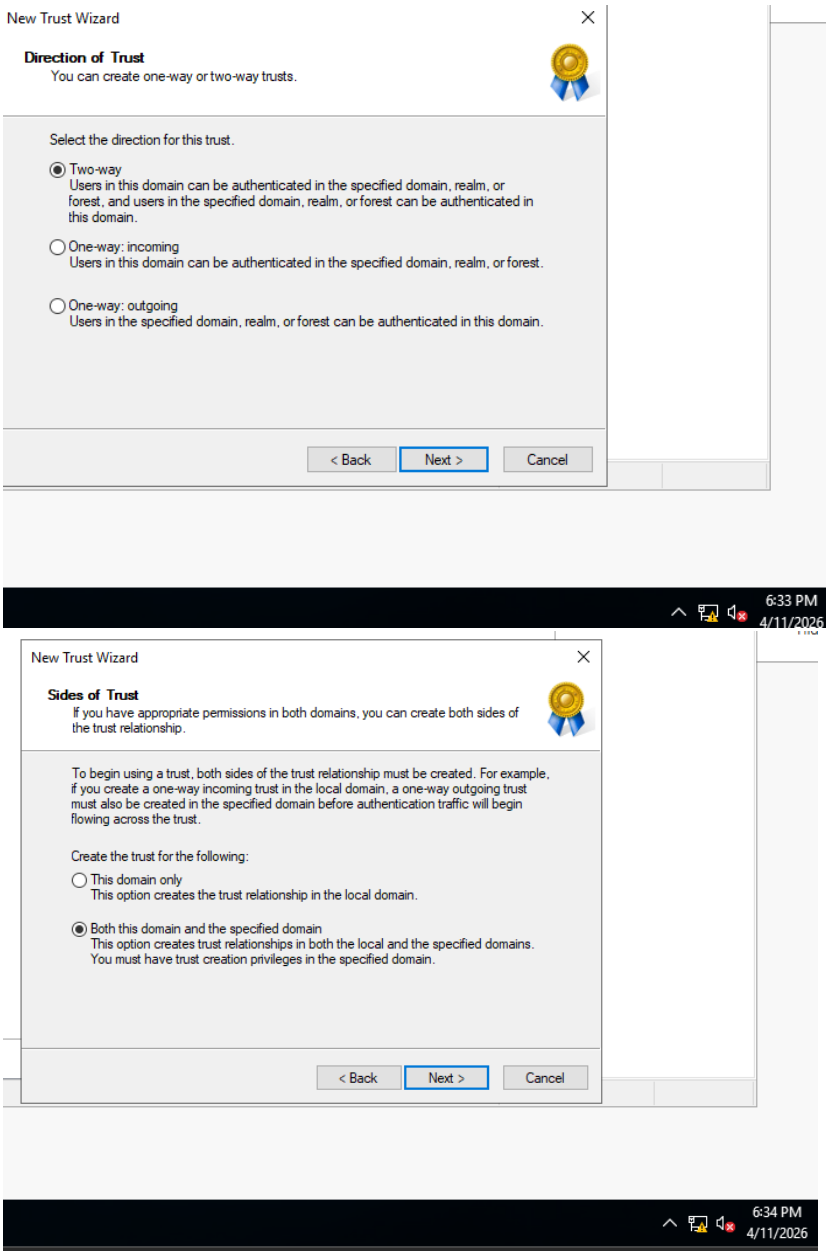
Section 2 – Configuration du trust (10 pts)

1. Configurez un **two-way forest trust** entre contoso.com et fabrikam.com.
   1.1 commencer par ajouter le IP de DC-fabrikam(192.168.151.11) dans le DNS de contoso :



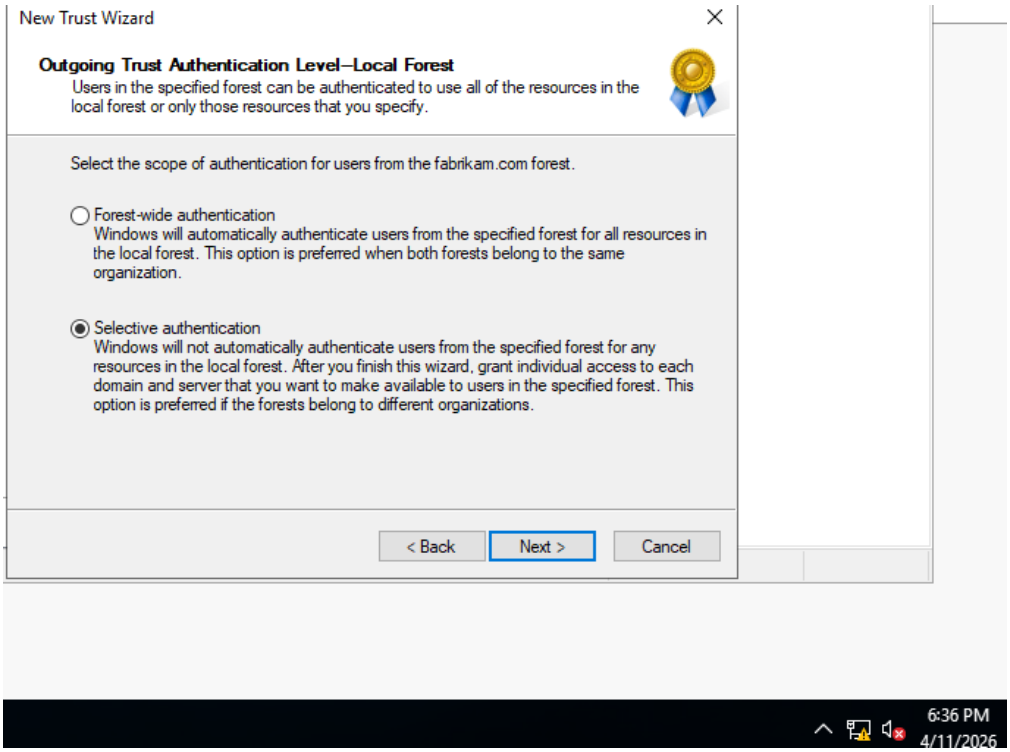

Active Directory Domains and Trusts
File Action View Help
Active Directory Domains and Trust
contosoYL.com
Name
Type
There are no items to show in this view.
Actions
contosoYL.com
More Actions
New Trust Wizard
Trust Name
You can create a trust by using a NetBIOS or DNS name.
Type the name of the domain, forest, or realm for this trust. If you type the name of a forest, you must type a DNS name.
Example NetBIOS name: supplier01-int
Example DNS name: supplier01-internal.microsoft.com
Name:
fabrikam.com
< Back
Next >
Cancel
Saturday, April 11,
6:26 PM
4/11/2026


New Trust Wizard
Trust Type
This domain is a forest root domain. If the specified domain qualifies, you can create a forest trust.
Select the type of trust you want to create.
External trust
An external trust is a nontransitive trust between a domain and another domain outside the forest. A nontransitive trust is bounded by the domains in the relationship.
Forest trust
A forest trust is a transitive trust between two forests that allows users in any of the domains in one forest to be authenticated in any of the domains in the other forest.
< Back
Next >
Cancel
6:33 PM

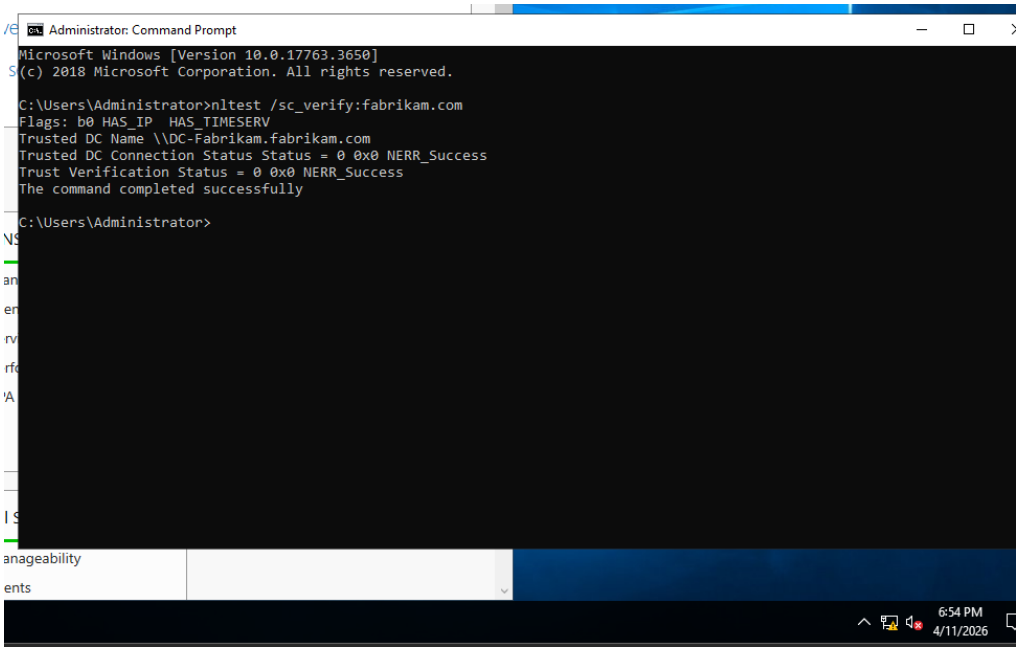


○ Option : **Forest trust** (transitive dans chaque forêt).

○ Choisissez **Selective authentication** (sécurité renforcée).



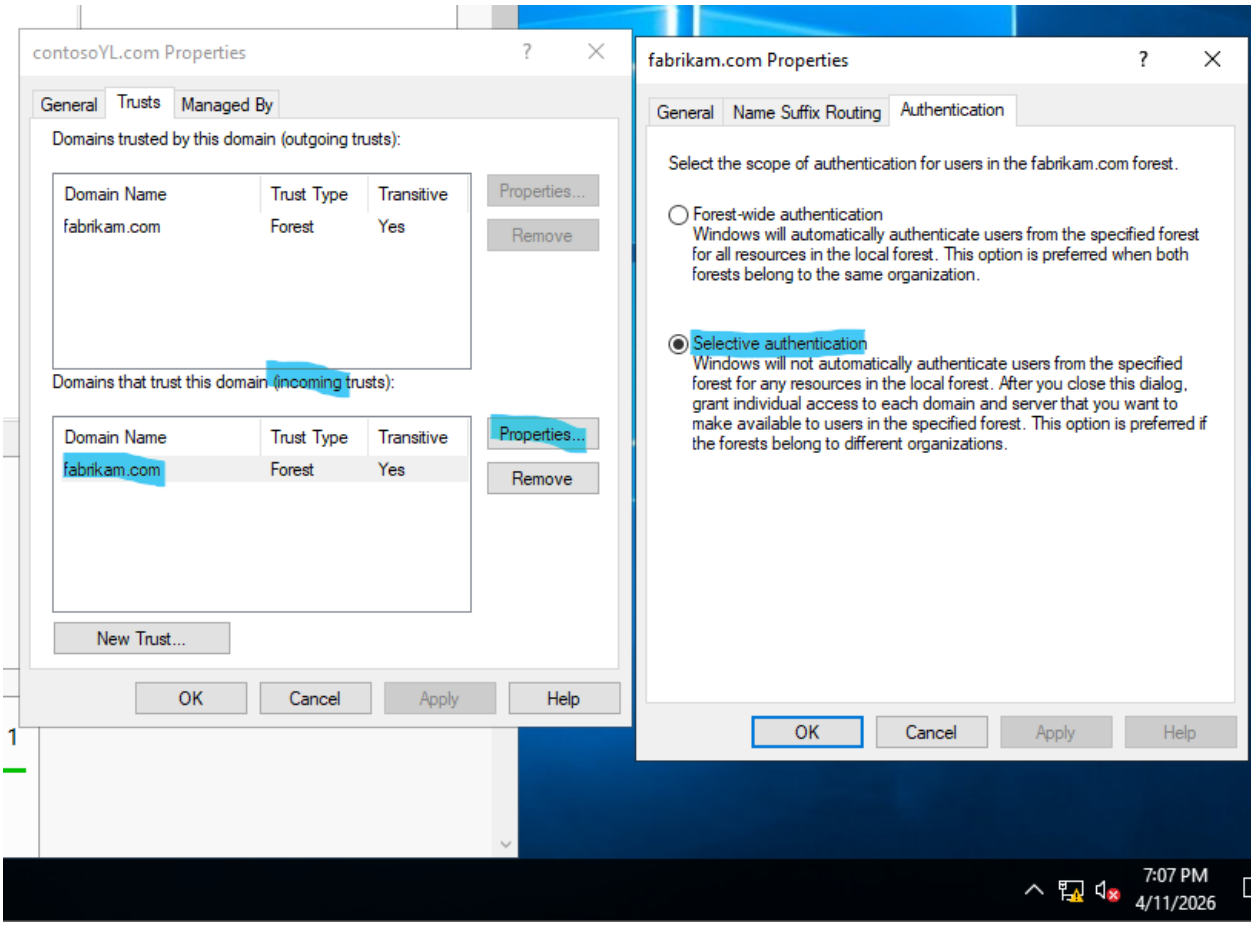
2. Validez le trust (Netdom /nltest /sc_verify). (à quoi servent ces commandes?)



nltest /sc_ verify : Vérifie le secure channel qui est le canal chiffré entre les deux DC qui sert à l'authentification Kerberos entre les forêts

netdom trust /verify : Vérifie que le trust existe et que le **mot de passe du trust** est bien synchronisé entre les deux DC

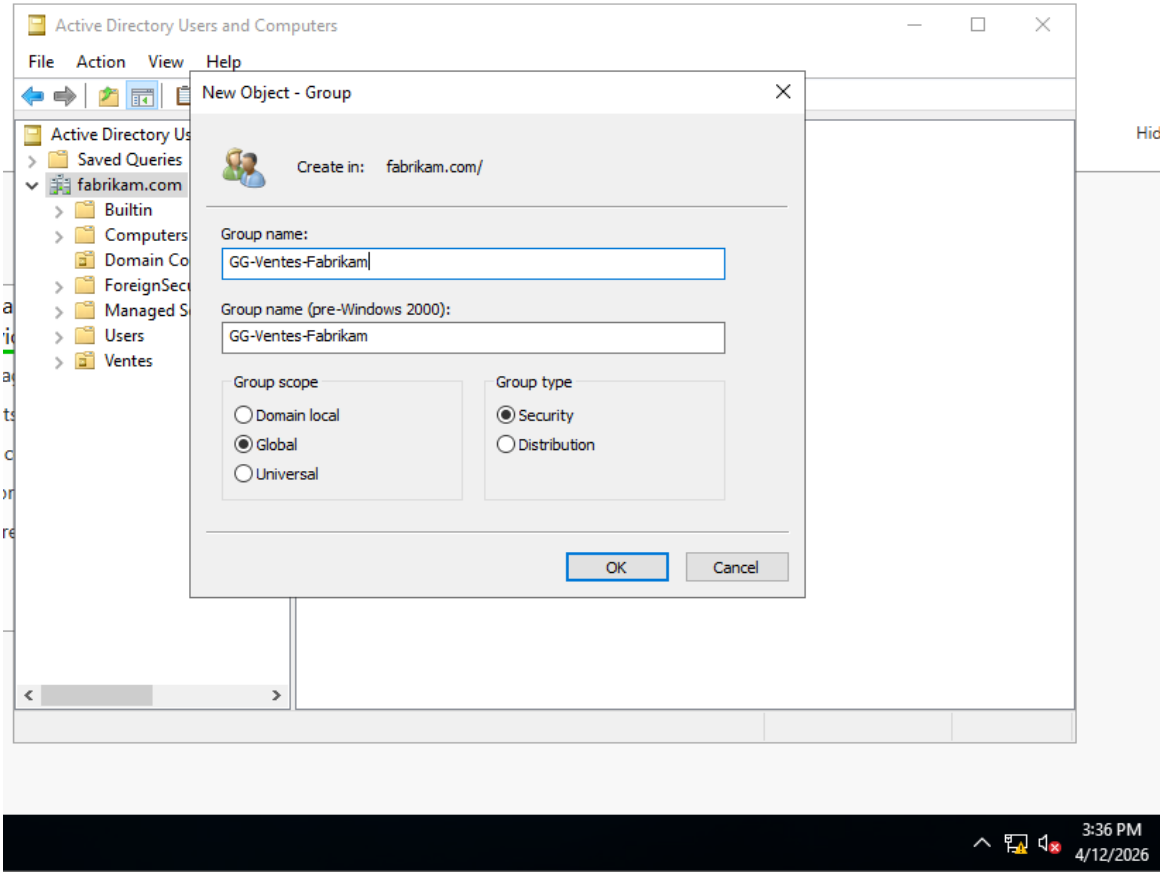
3. Activez l'authentification sélective sur le domaine contoso.com pour le trust entrant depuis fabrikam.com.



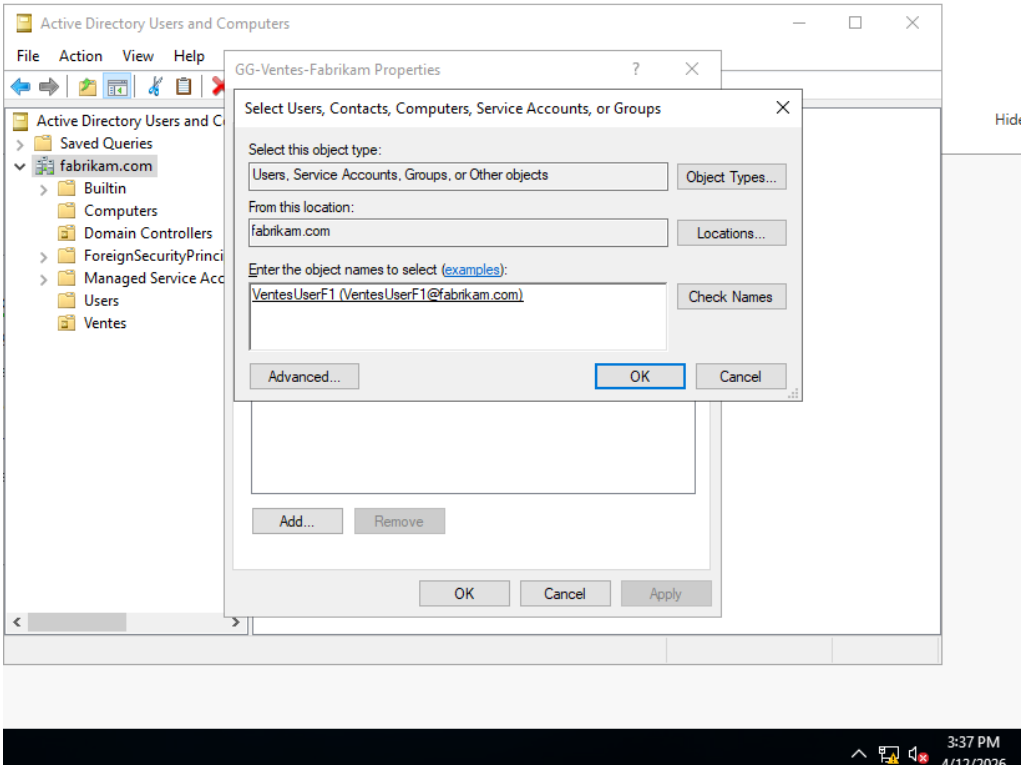
Section 3 – Groupes et nesting (AGDLP / AGUDLP) (10 pts)

1. Dans **fabrikam.com** (domaine des utilisateurs) :

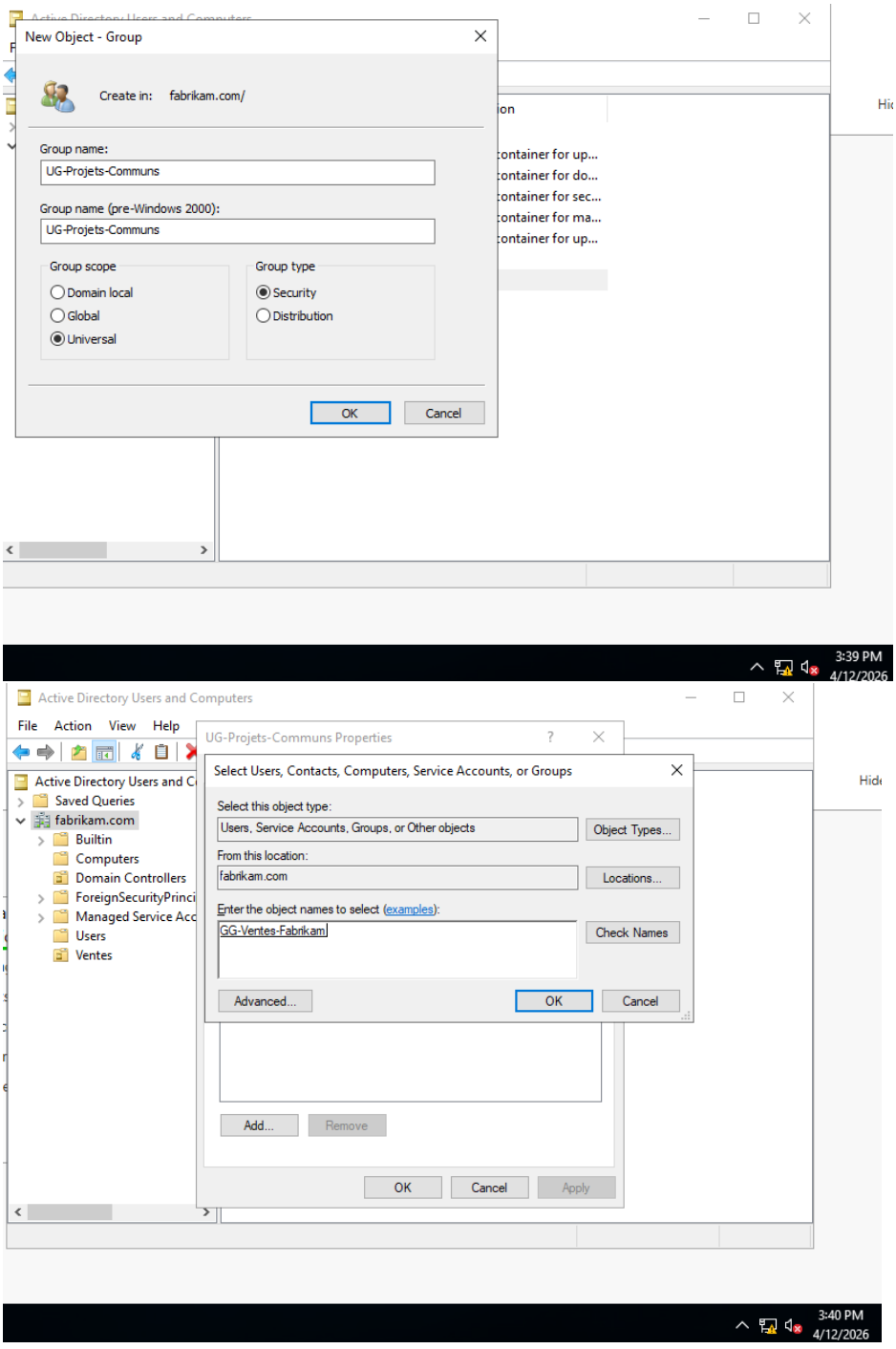
○ Créez un **Global Group** : GG-Ventes-Fabrikam :



○ **Ajoutez VentesUserF1 dedans :**

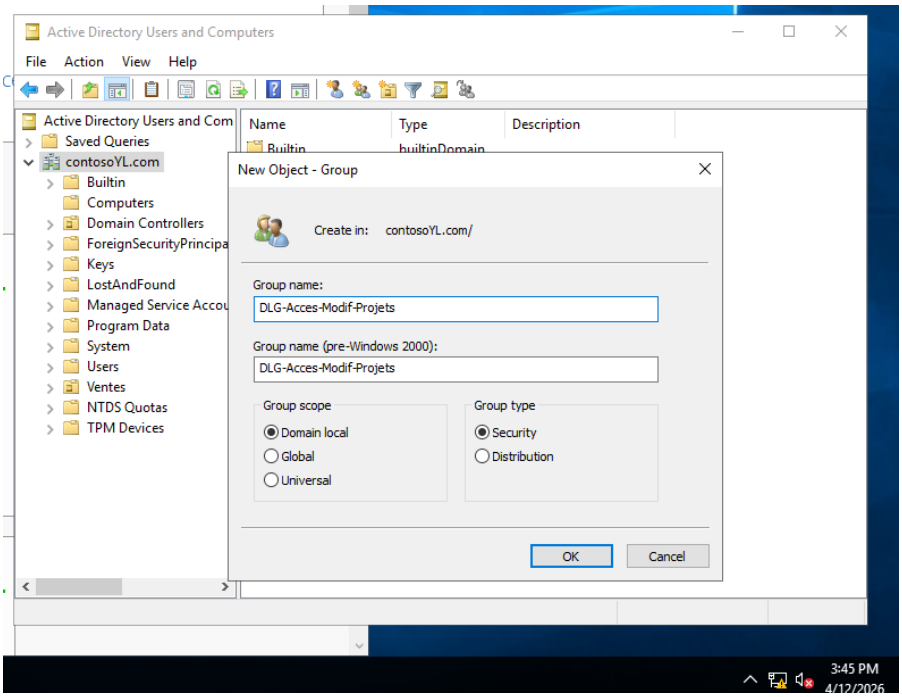


3. Créez un **Universal Group** : UG-Projets-Communs dans contoso.com ou fabrikam.com et ajoutez-y le GG-Ventes-Fabrikam.

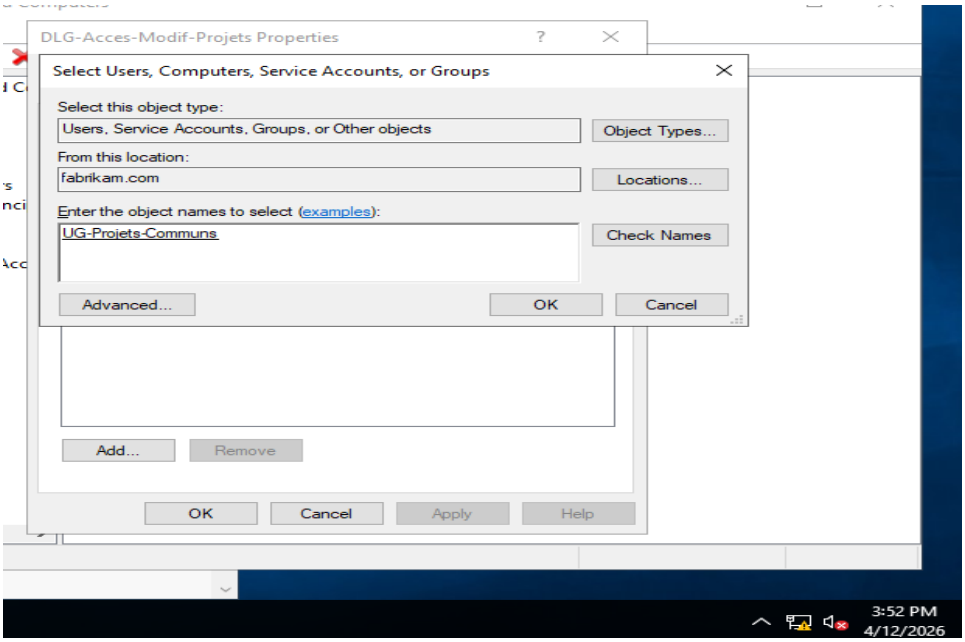


3. Dans **contoso.com** (domaine des ressources) :

○ Créez un **Domain Local Group** : DLG-Acces-Modif-Projets

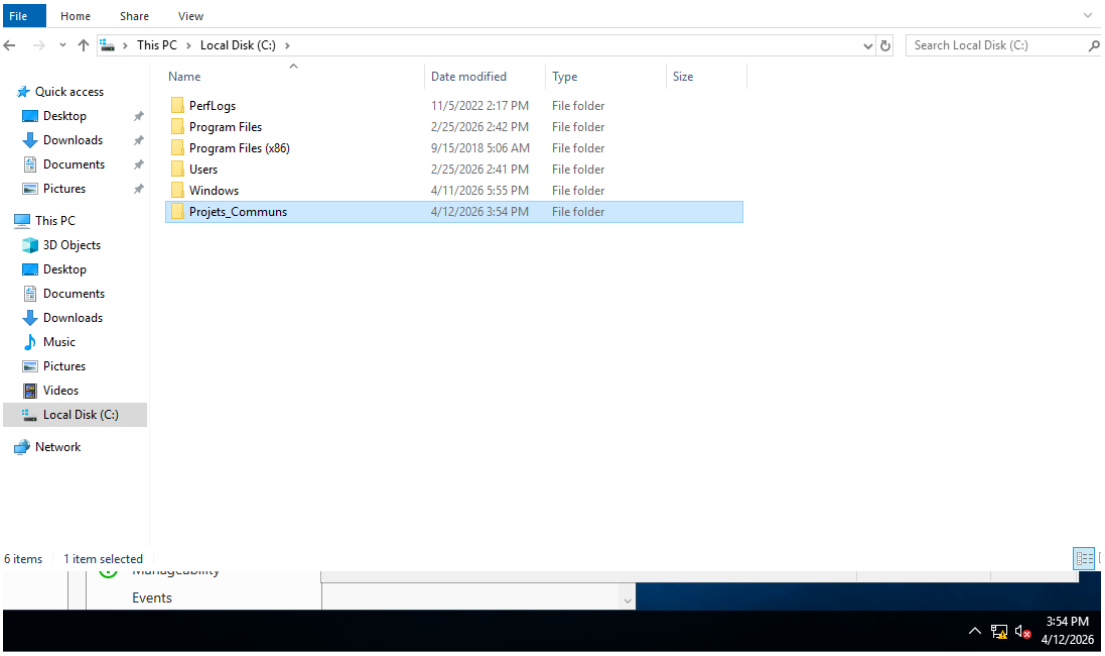


○ Ajoutez-y le GG-Ventes-Fabrikam (ou UG si créé).

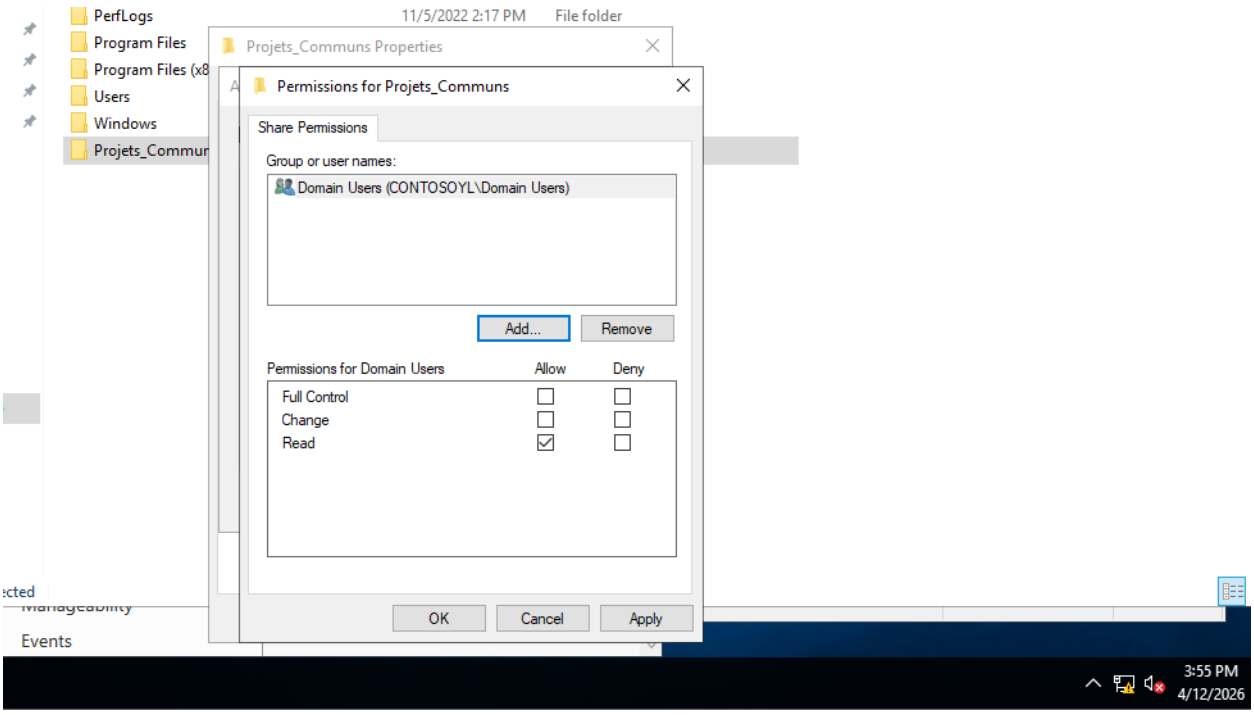


Section 4 – Permissions NTFS et partage (10 pts) Sur le FileServer de Contoso :

1. Créez un dossier C:\Projets_Communs.

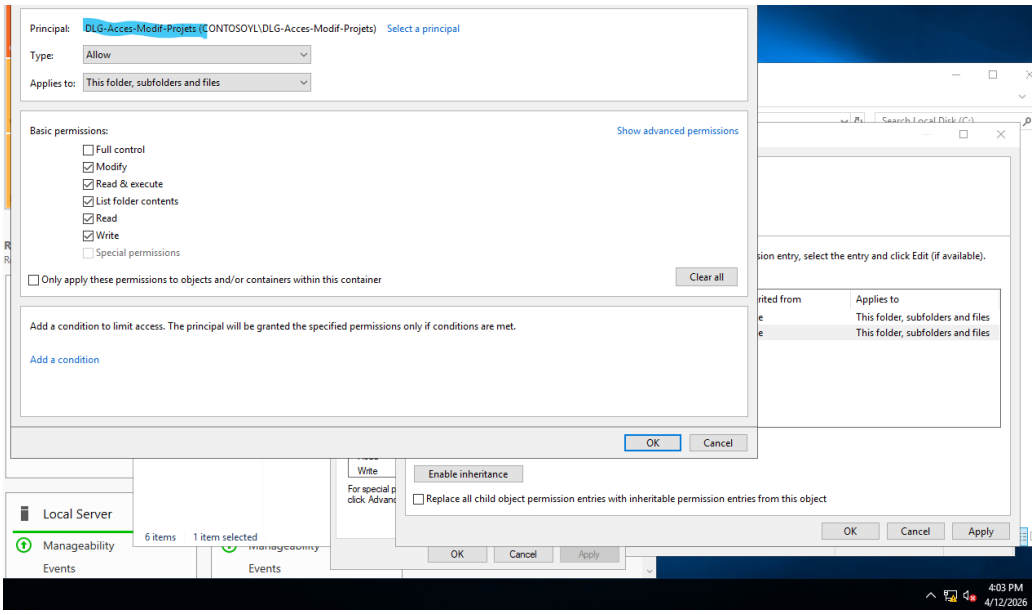


2. Partagez-le: Projets_Communs (share permissions : Domain users = Read).

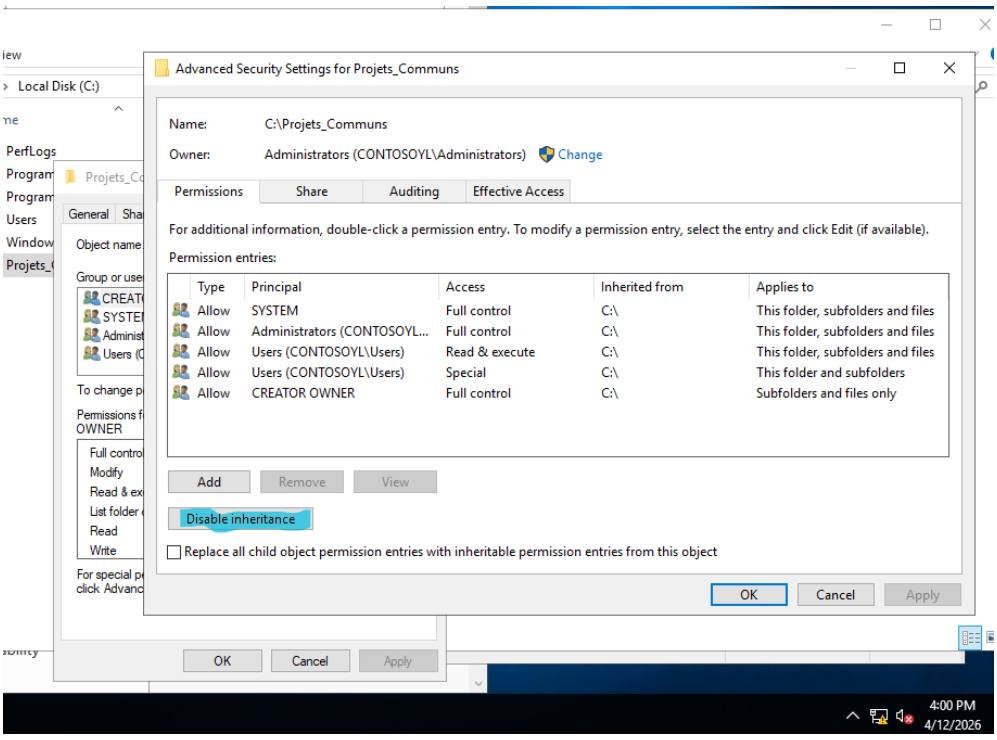


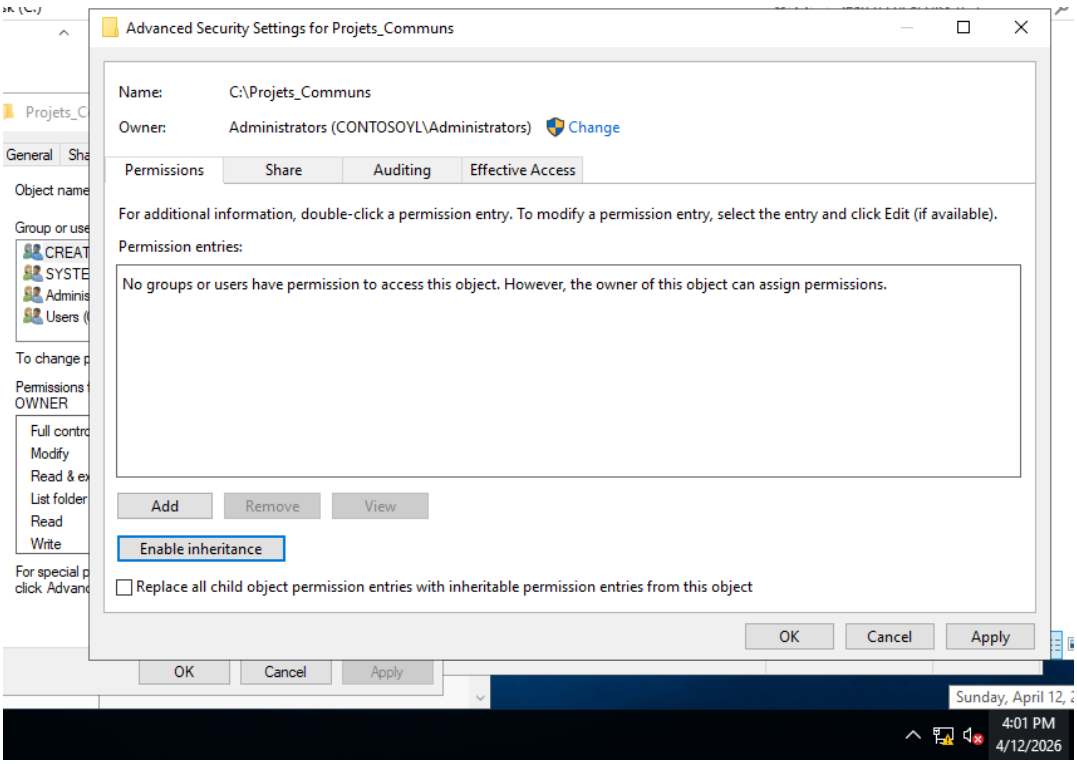
3. Configurez les **NTFS permissions** :

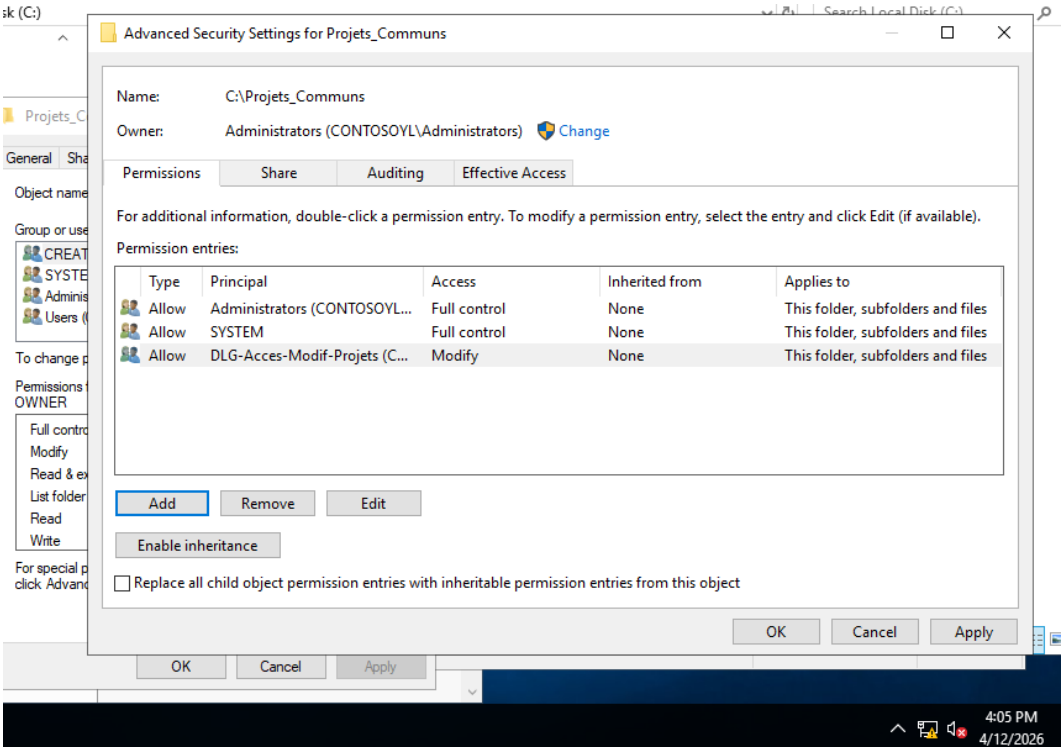
○ **DLG-Acces-Modif-Projets → Modify (ou Read/Write/Execute + List folder contents).**



○ **Supprimez les permissions par défaut sauf Administrators et SYSTEM.**




Advanced Security Settings for Projets_Communs
Name: C:\Projets_Communs
Owner: Administrators (CONTOSOYL\Administrators) Change
Permissions Share Auditing Effective Access
For additional information, double-click a permission entry. To modify a permission entry, select the entry and click Edit (if available).
Permission entries:
No groups or users have permission to access this object. However, the owner of this object can assign permissions.
Add Remove View
Enable inheritance
Replace all child object permission entries with inheritable permission entries from this object
OK Cancel Apply
4:01 PM
4/12/2026


Advanced Security Settings for Projets_Communs
Name: C:\Projets_Communs
Owner: Administrators (CONTOSOYL\Administrators) Change
Permissions Share Auditing Effective Access
For additional information, double-click a permission entry. To modify a permission entry, select the entry and click Edit (if available).
Permission entries:
Type Principal Access Inherited from Applies to
Allow Administrators (CONTOSOYL... Full control None This folder, subfolders and files
Allow SYSTEM Full control None This folder, subfolders and files
Allow DLG-Acces-Modif-Projets (C... Modify None This folder, subfolders and files
Add Remove Edit
Enable inheritance
Replace all child object permission entries with inheritable permission entries from this object
OK Cancel Apply
4:05 PM
4/12/2026

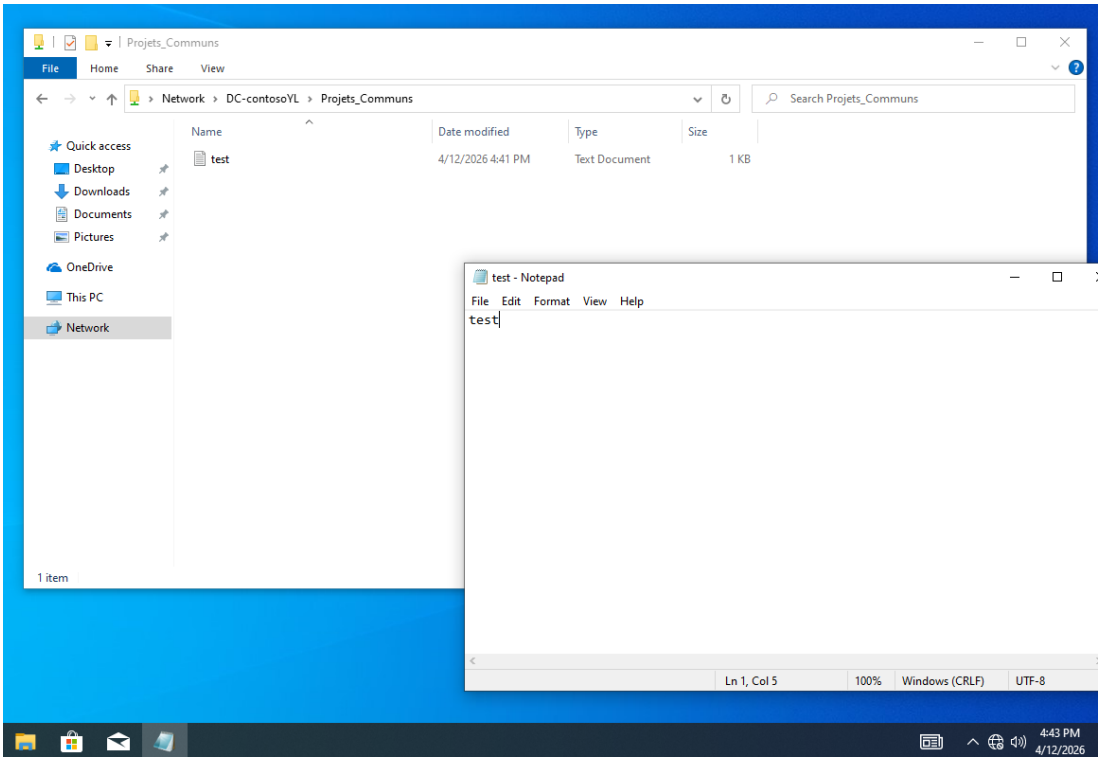
Section 5 – Tests et dépannage (10 pts)

1. Depuis le client fabrikam.com, connectez-vous avec VentesUserF1.

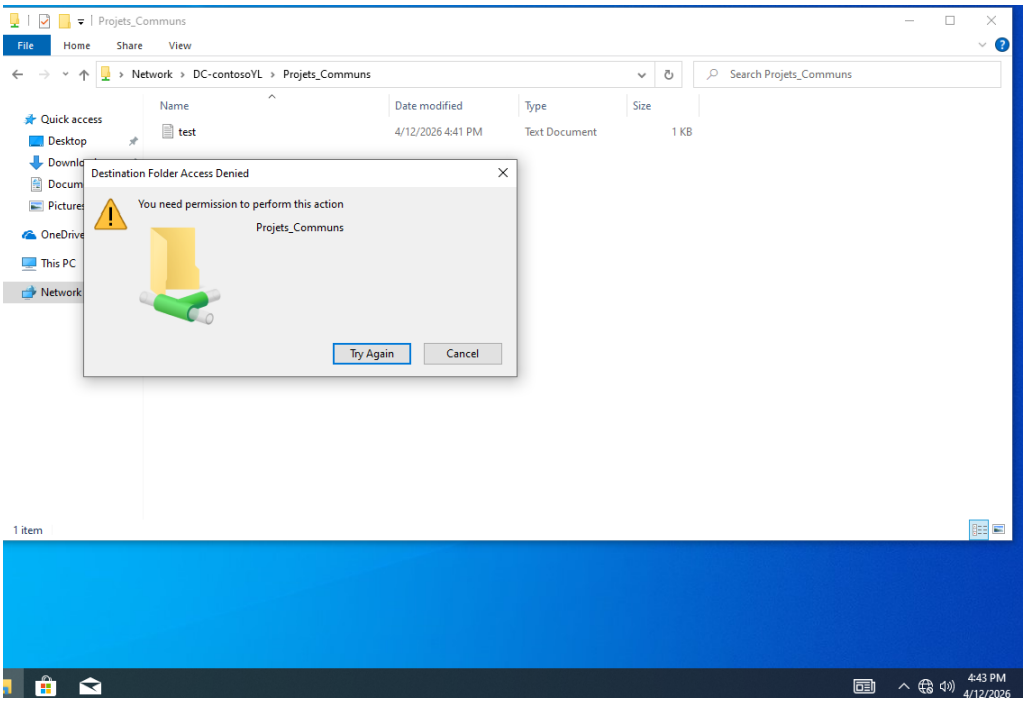
2. Essayez d'accéder à Projets_Communs.

○ Testez : lecture, création fichier, modification, suppression.

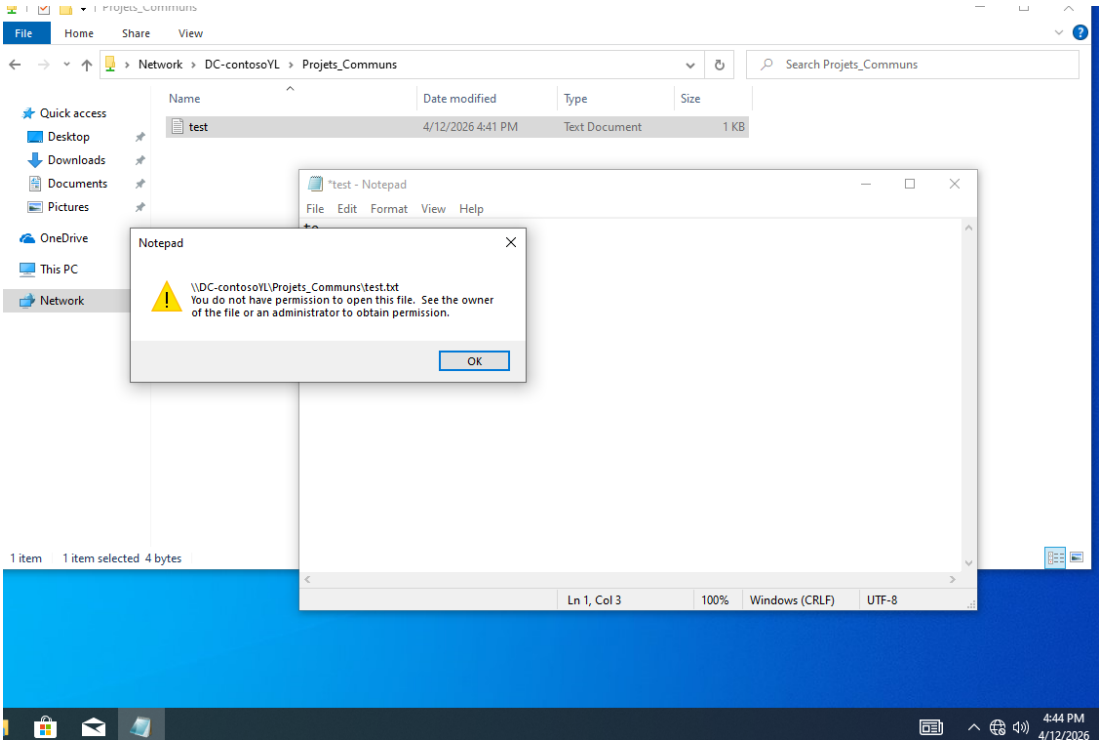
1.lecture :



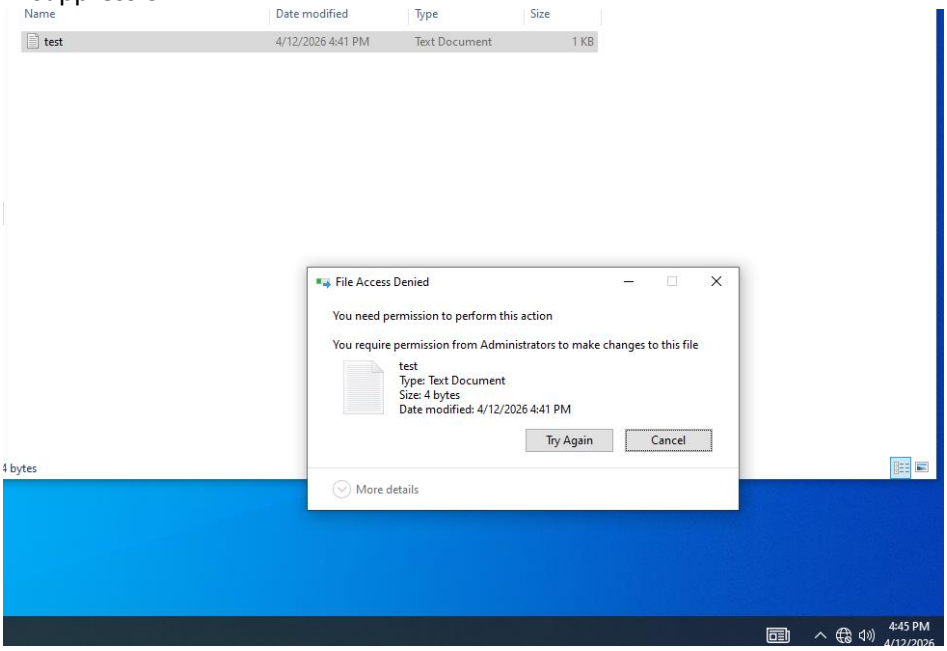
2. création de fichie :



3. modification :

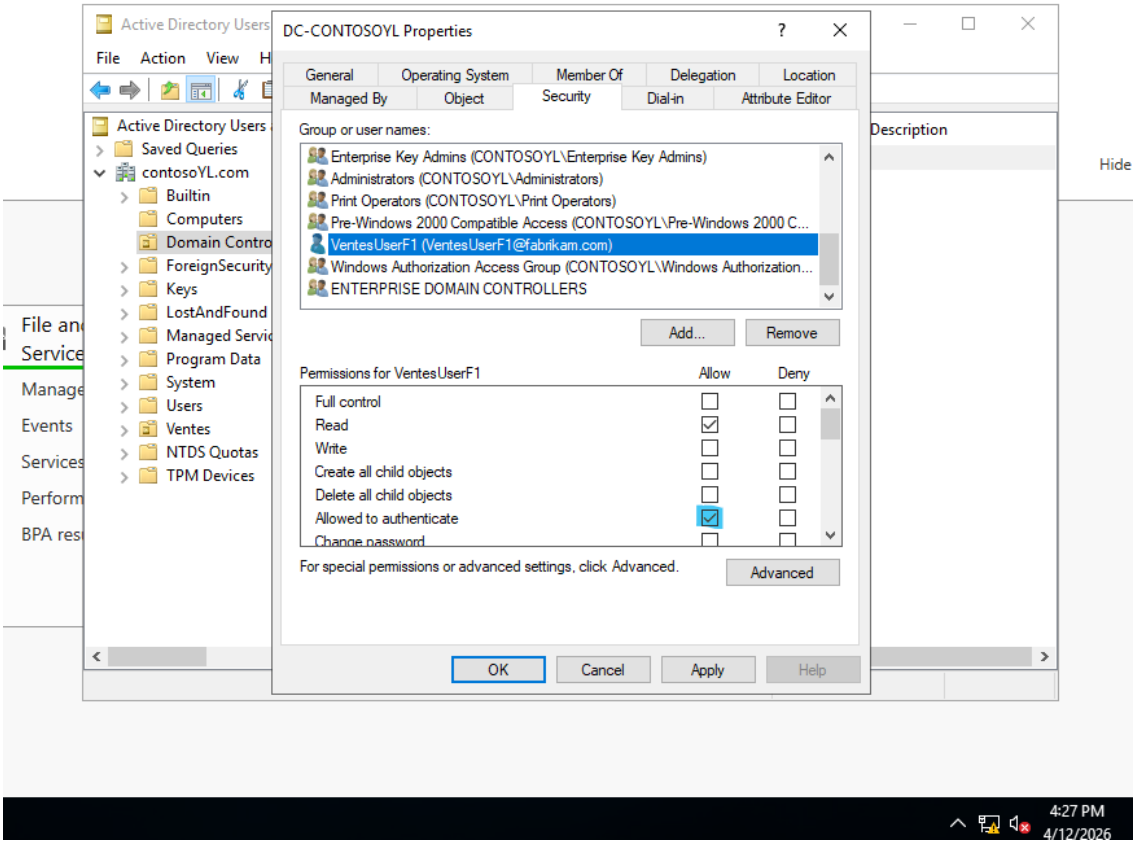


4. suppression :



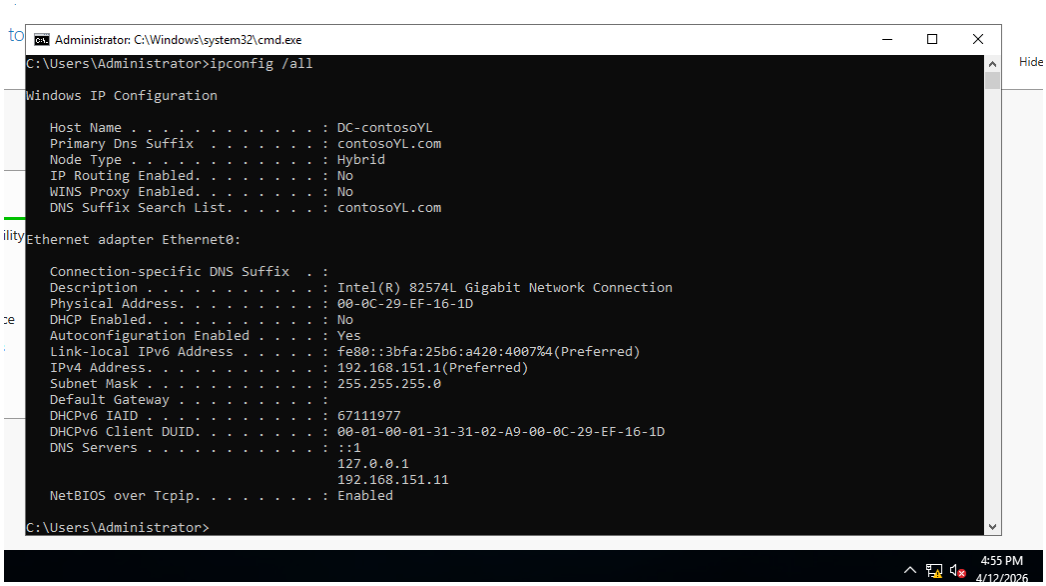
3. Si accès refusé → activez l'authentification pour le compte sur le trust (Active Directory Domains and Trusts → Properties → Authentication tab → Allow).

- J'ai ajouté le user VentesUserF1 au DC-ContosoYL en lui donnant le droit d'être authentifié :



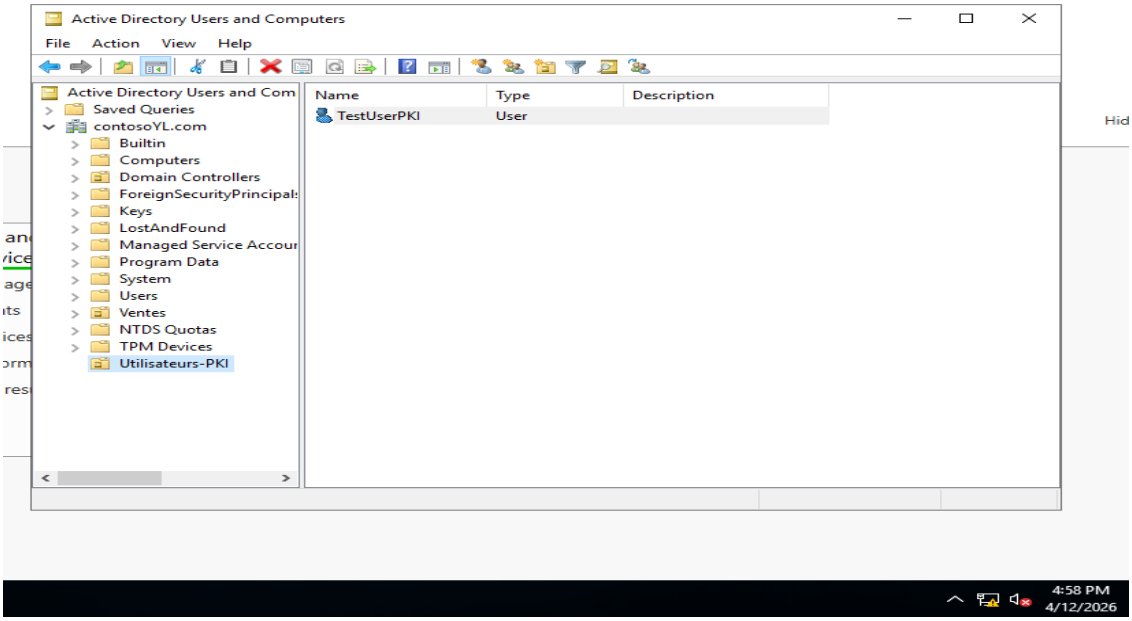
Section 6 – Préparation CA (10 pts)

1. Assurez-vous que le serveur CA (voir TP1 pour savoir sur quel serveur l'installer) est joint au domaine et a une IP statique.

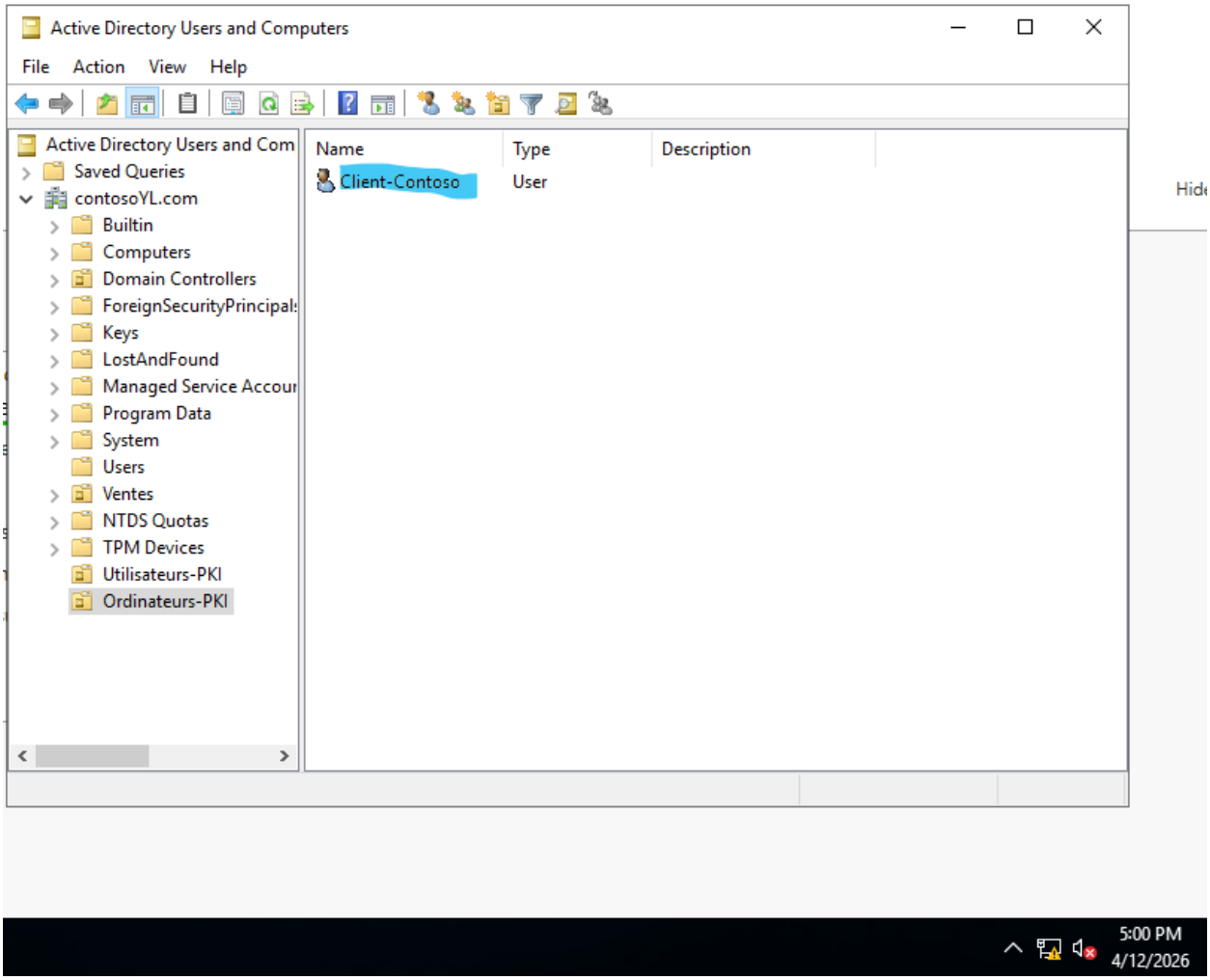


2. Créez des OU et utilisateurs/tests :

- OU = Utilisateurs-PKI → utilisateur TestUserPKI



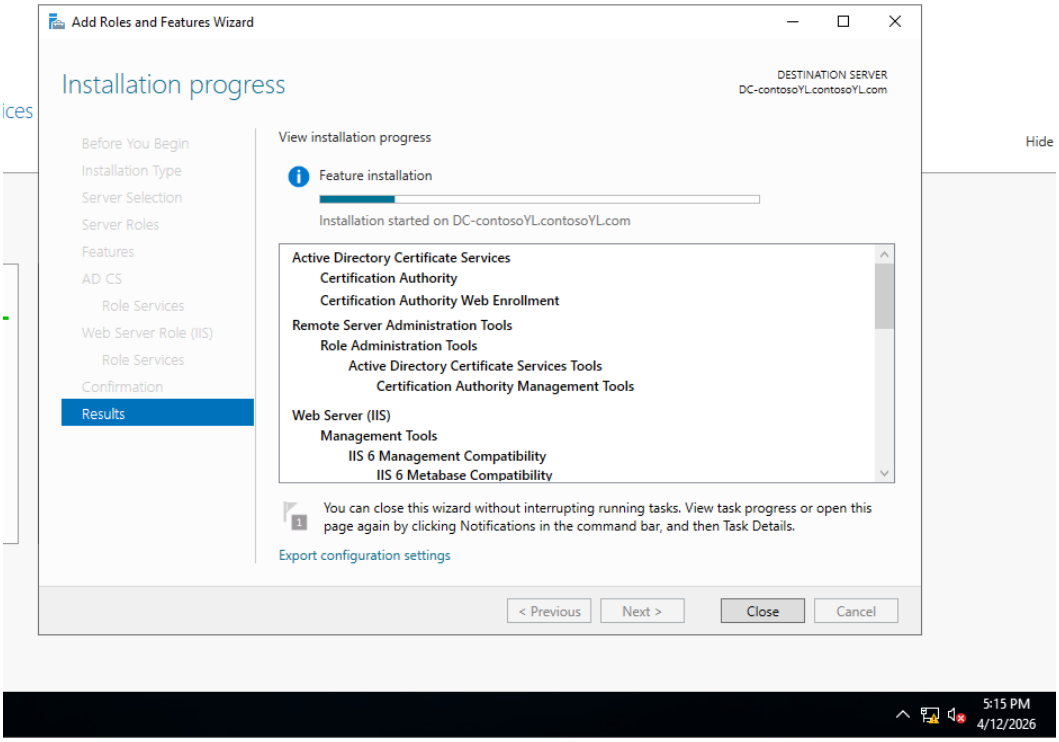
- OU = Ordinateurs-PKI → (le client sera dedans)



Section 7 – Installation et configuration Enterprise CA (10 pts)

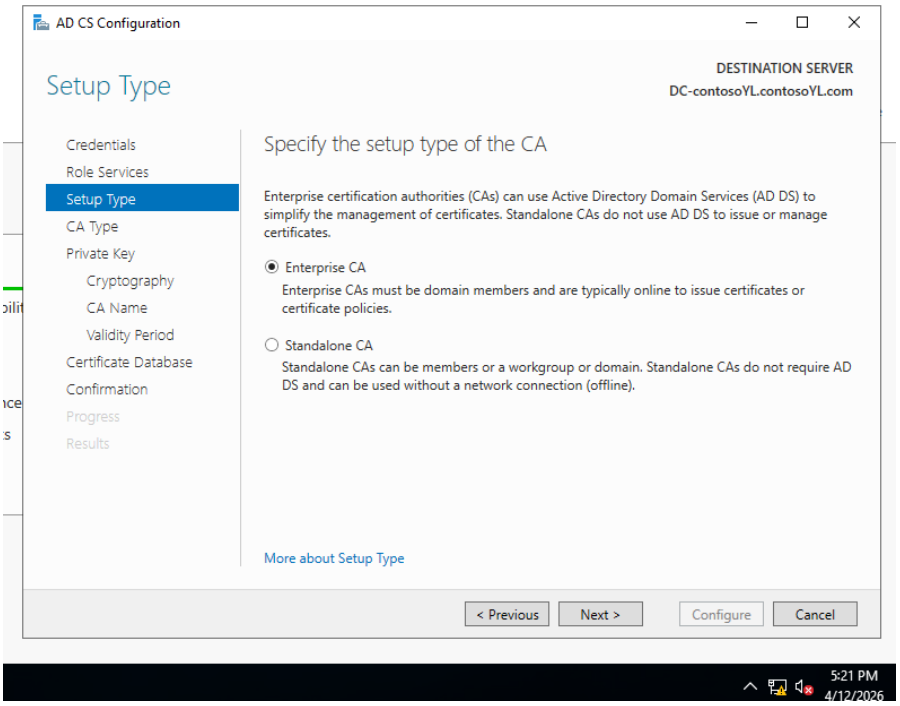
1. Ajoutez le rôle Active Directory Certificate Services via Server Manager.

- Sélectionnez : Certification Authority + Certification Authority Web Enrollment

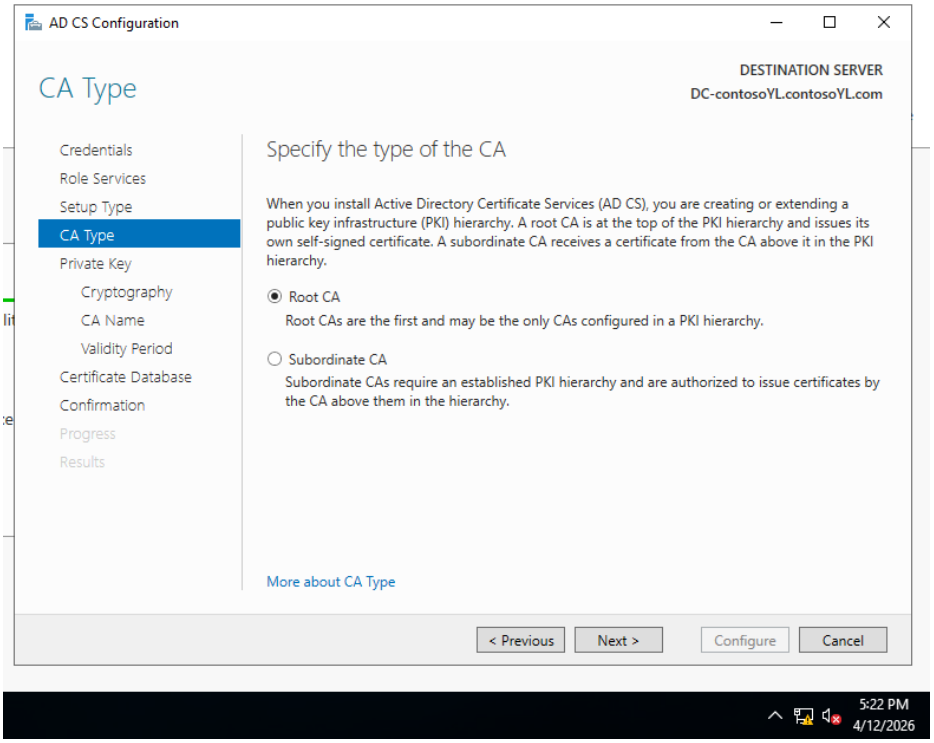


2. Configurez :

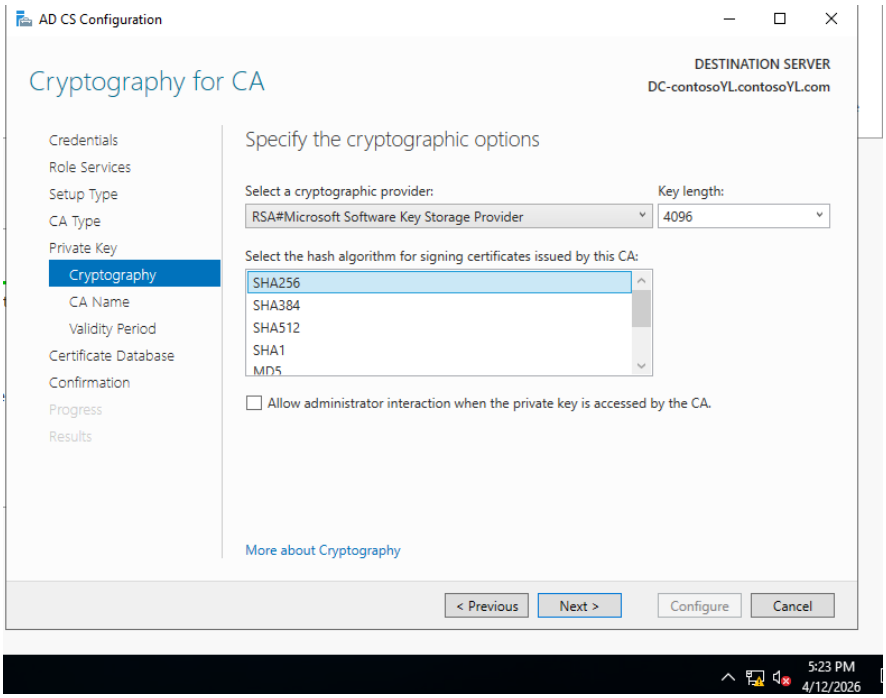
- Enterprise CA (intègre AD).



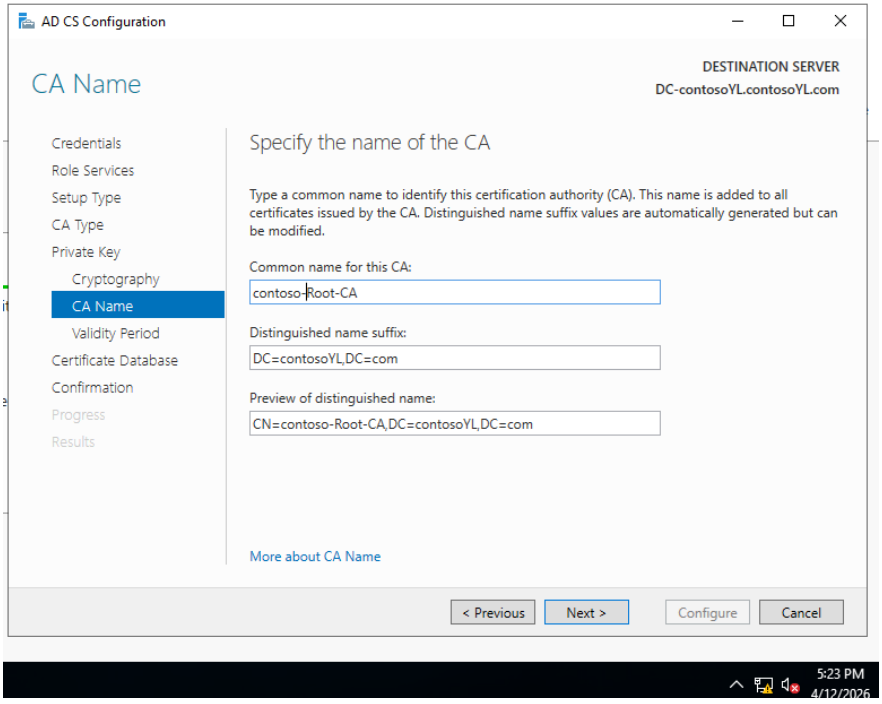
- Root CA** (en lab ; en prod, Subordinate).



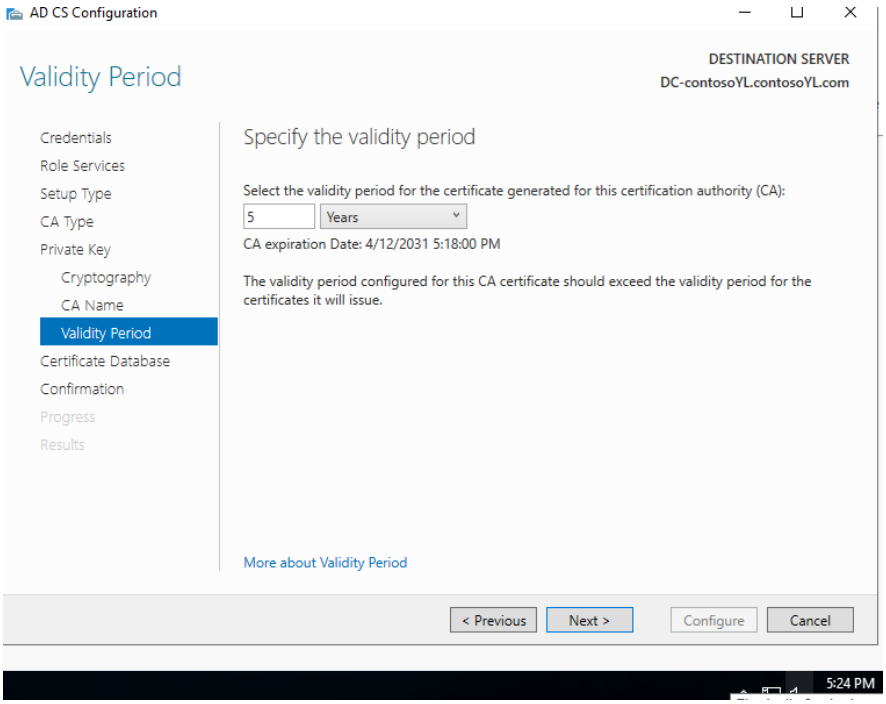
- Cryptographie : RSA 4096 bits, SHA-256.



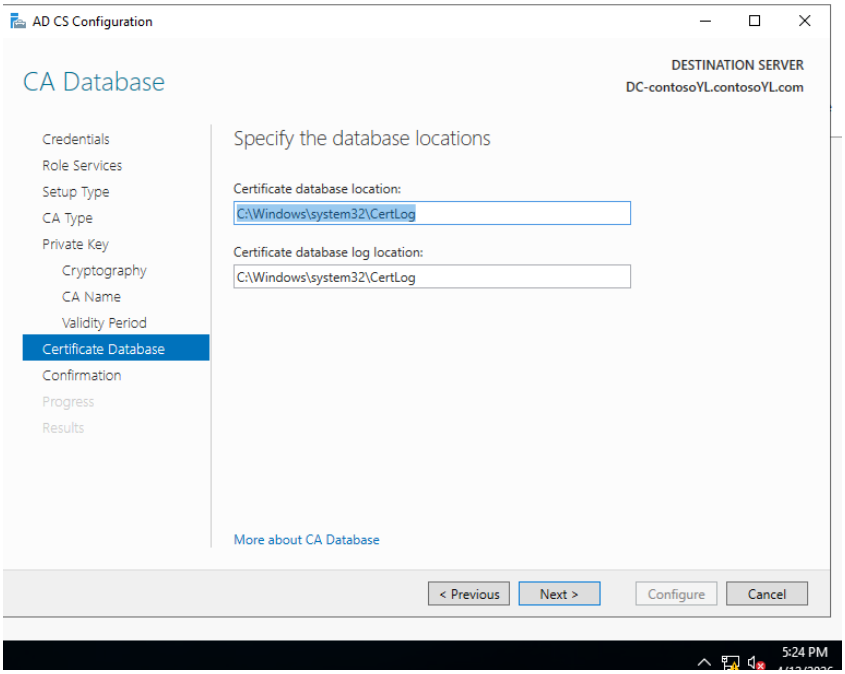
- Nom CA : Contoso-Root-CA



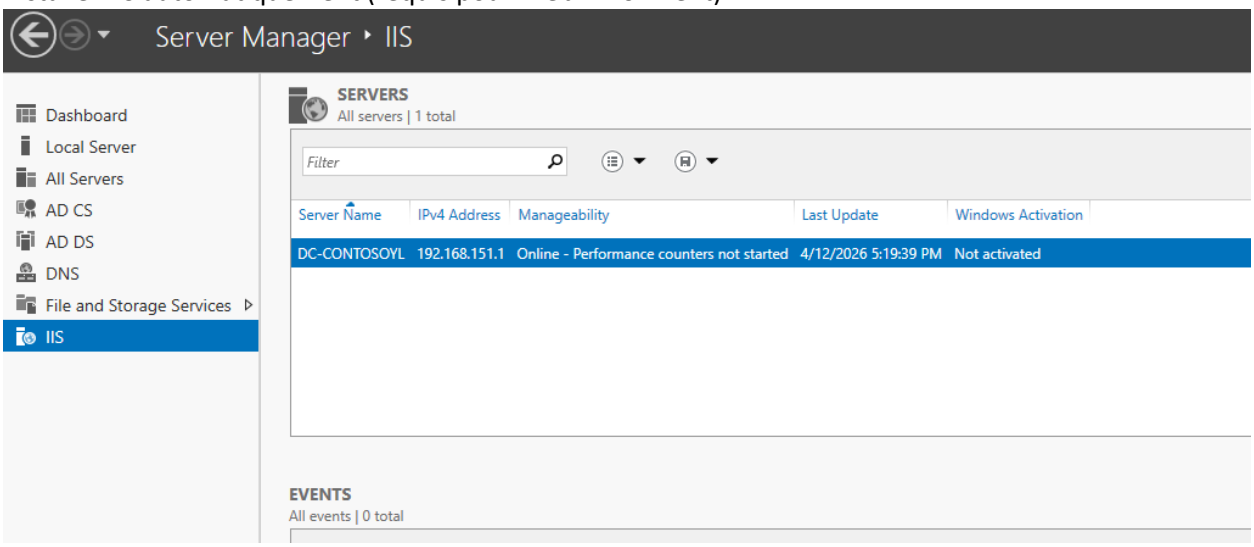
- Validité : 10 ans (ou 5 ans).



- Database et logs : emplacements par défaut.



2. Installez IIS automatiquement (requis pour Web Enrollment).

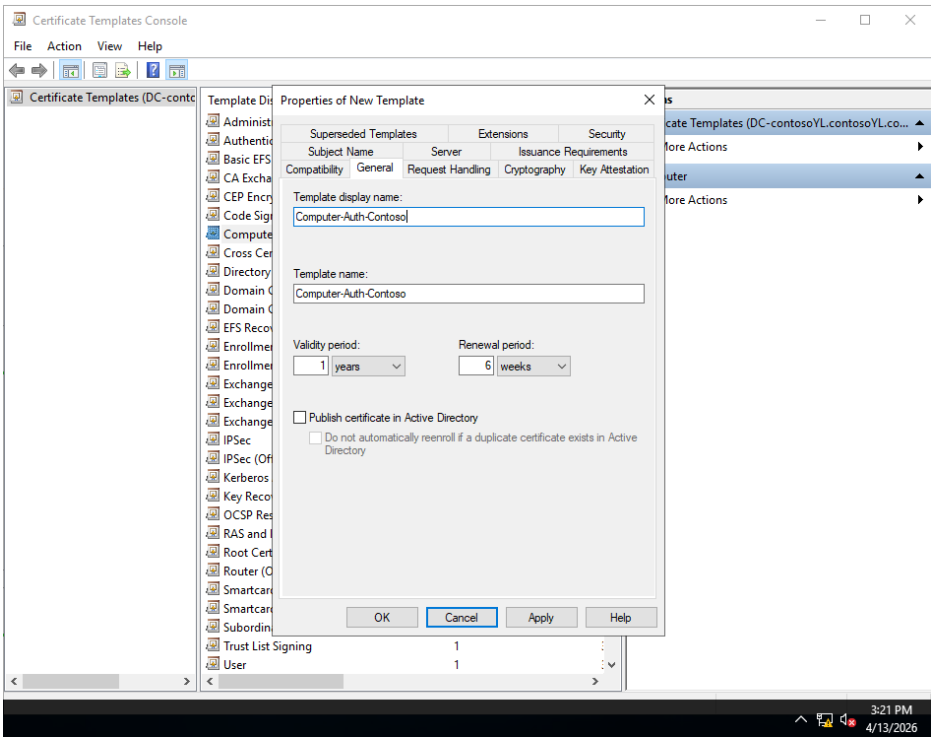


Section 8 – Templates + Auto-Enrollment (10 pts)

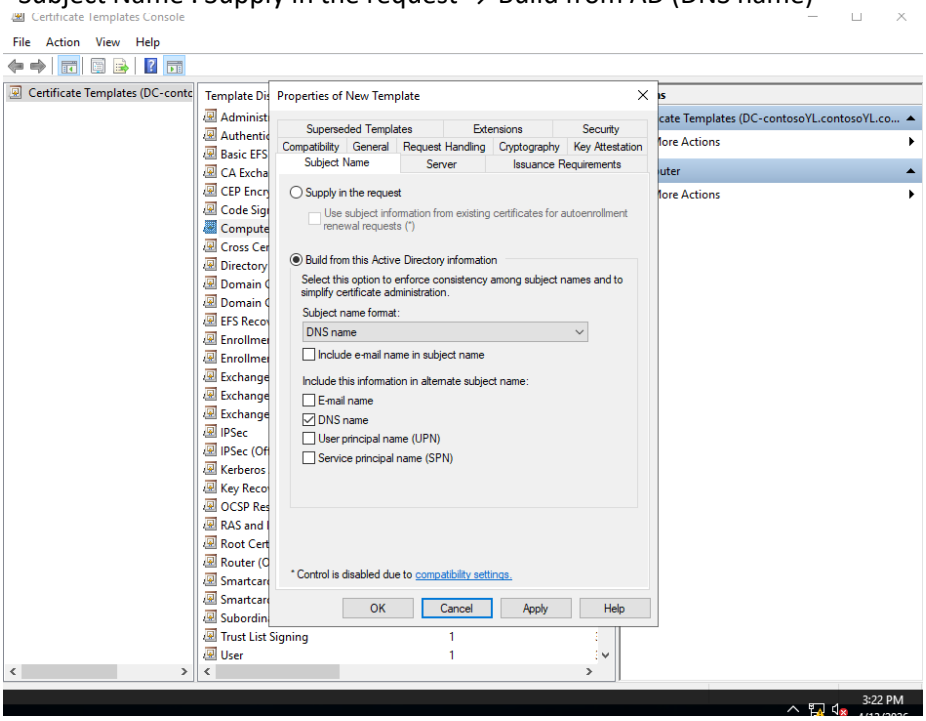
1. Ouvrez Certification Authority console → clic droit Certificate Templates → Manage.

2. Dupliquer et configurez :

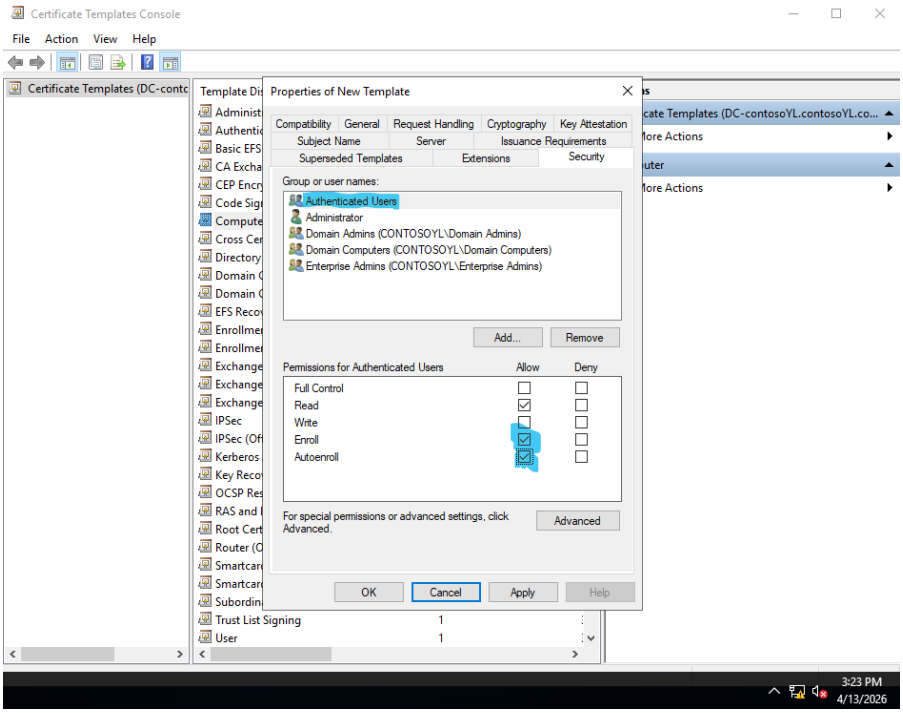
- Computer → nom : Computer-Auth-Contoso



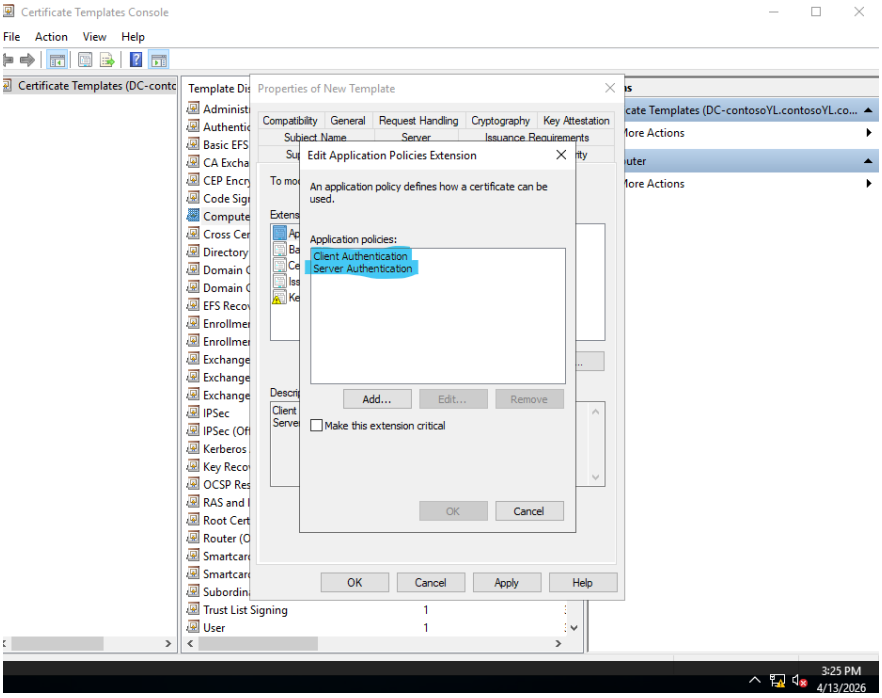
- Subject Name : Supply in the request → Build from AD (DNS name)



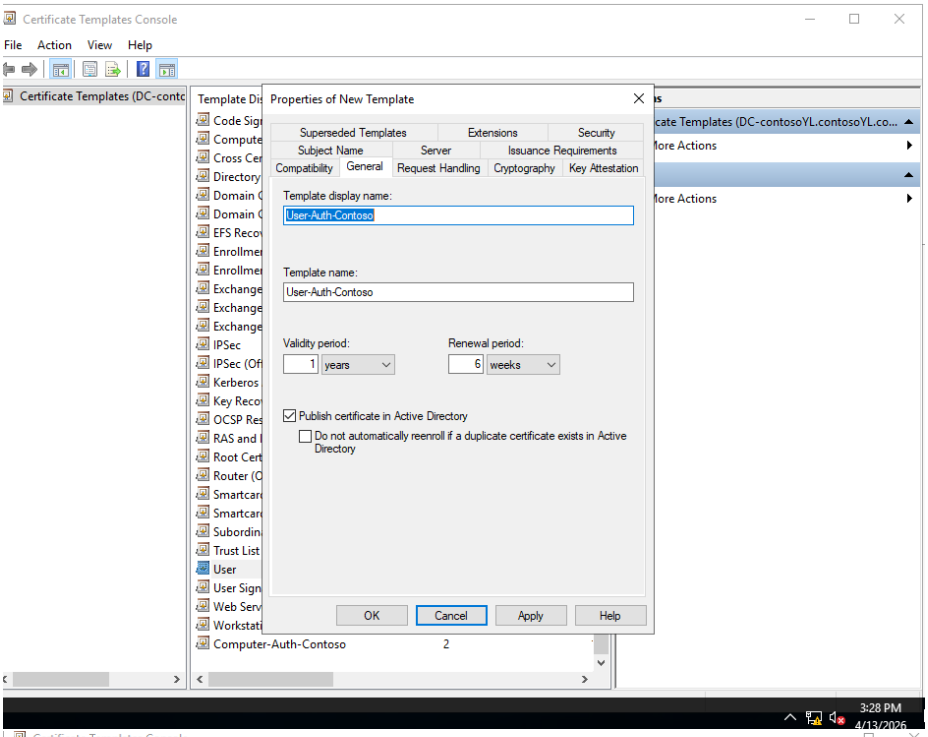
- Security : Authenticated Users → Enroll + Autoenroll

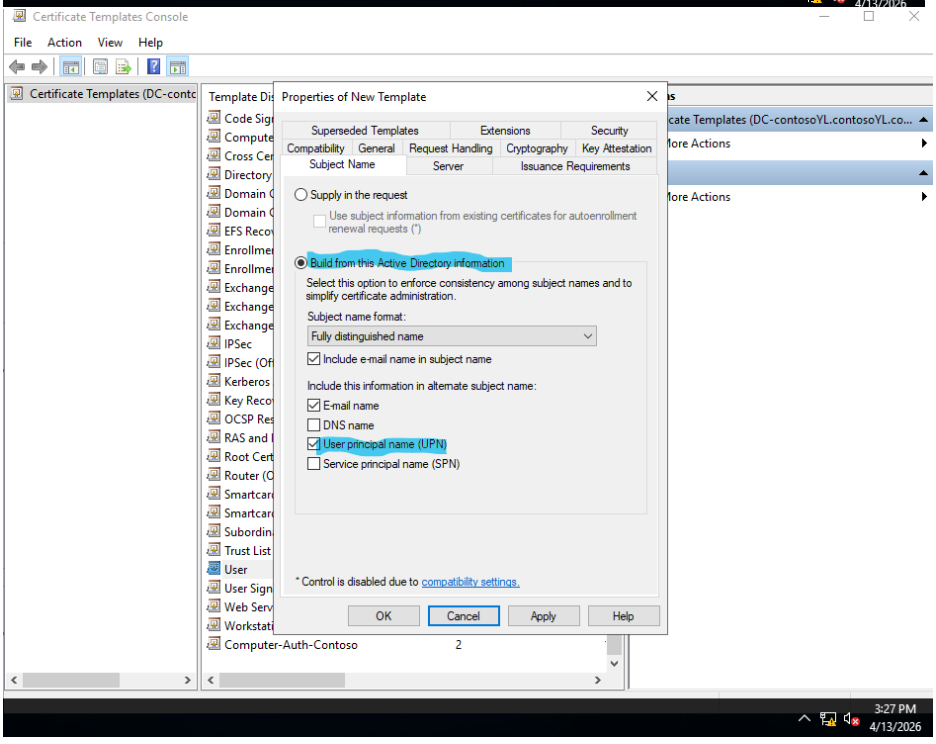


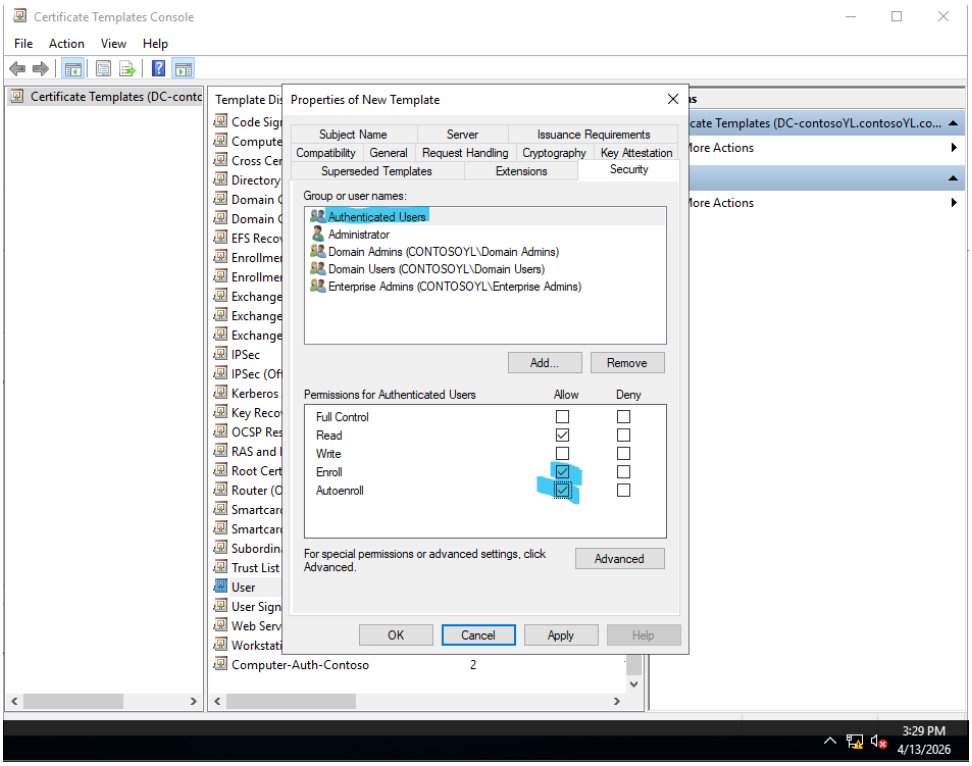
- Extensions : Client Authentication + Server Authentication

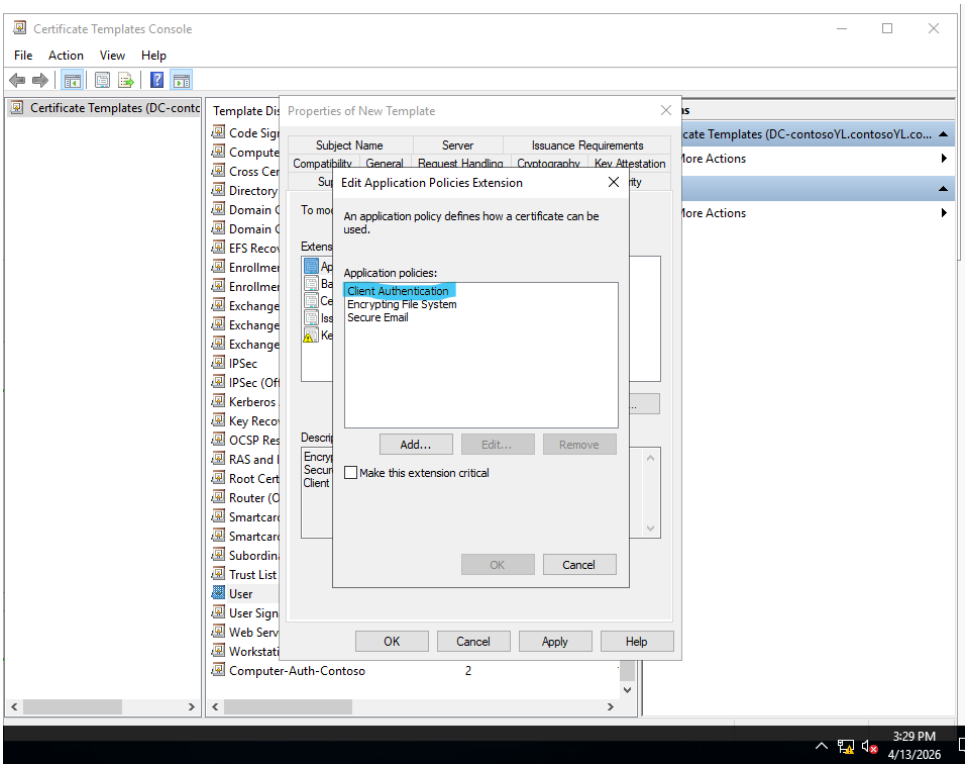


- User → nom : User-Auth-Contoso (similaire, pour authentification utilisateur)



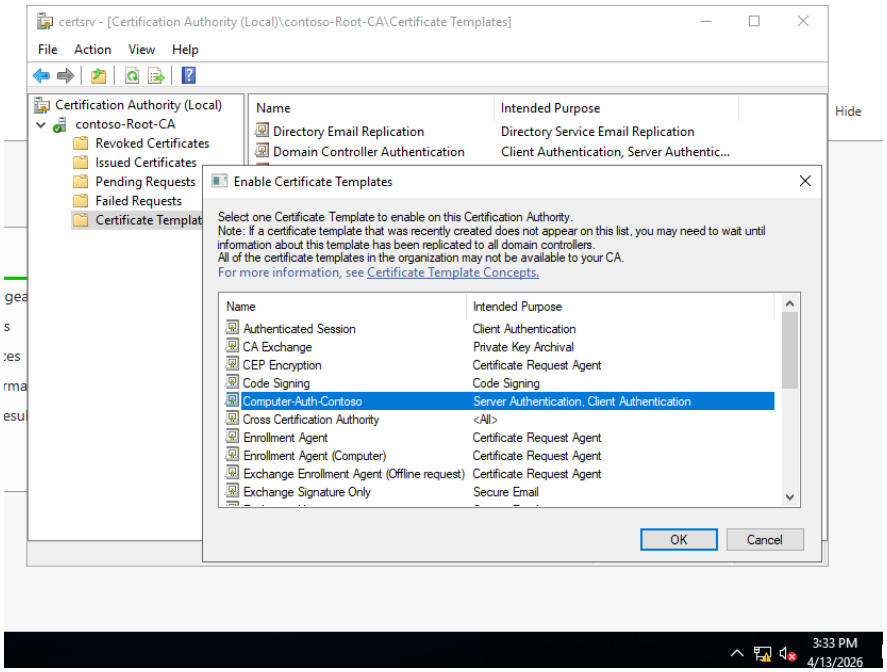


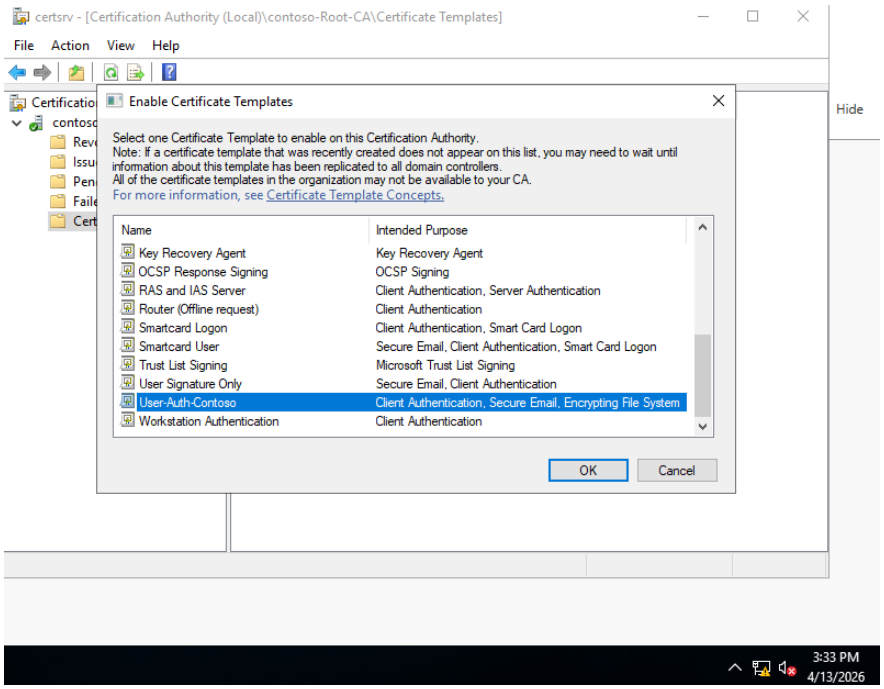

Certificate Templates Console
File Action View Help
Certificate Templates (DC-contc
Template Dis
Properties of New Template
Subject Name
Server
Issuance Requirements
Compatibility
General
Request Handling
Cryptography
Key Attestation
Superseded Templates
Extensions
Security
Group or user names:
Authenticated Users
Administrator
Domain Admins (CONTOSOYL\Domain Admins)
Domain Users (CONTOSOYL\Domain Users)
Enterprise Admins (CONTOSOYL\Enterprise Admins)
Add...
Remove
Permissions for Authenticated Users
Allow
Deny
Full Control
Read
Write
Enroll
Autoenroll
For special permissions or advanced settings, click Advanced.
Advanced
OK
Cancel
Apply
Help
Computer-Auth-Contoso
2
More Actions
3:29 PM
4/13/2026


Certificate Templates Console
File Action View Help
Certificate Templates (DC-contc
Template Dis
Properties of New Template
Subject Name
Server
Issuance Requirements
Edit Application Policies Extension
An application policy defines how a certificate can be used.
Application policies:
Client Authentication
Encrypting File System
Secure Email
Add...
Edit...
Remove
Make this extension critical
OK
Cancel
OK
Cancel
Apply
Help
Computer-Auth-Contoso
2
More Actions
3:29 PM
4/13/2026

3. Publiez les templates :

clic droit Certificate Templates → New → Certificate Template to Issue → sélectionnez-les.

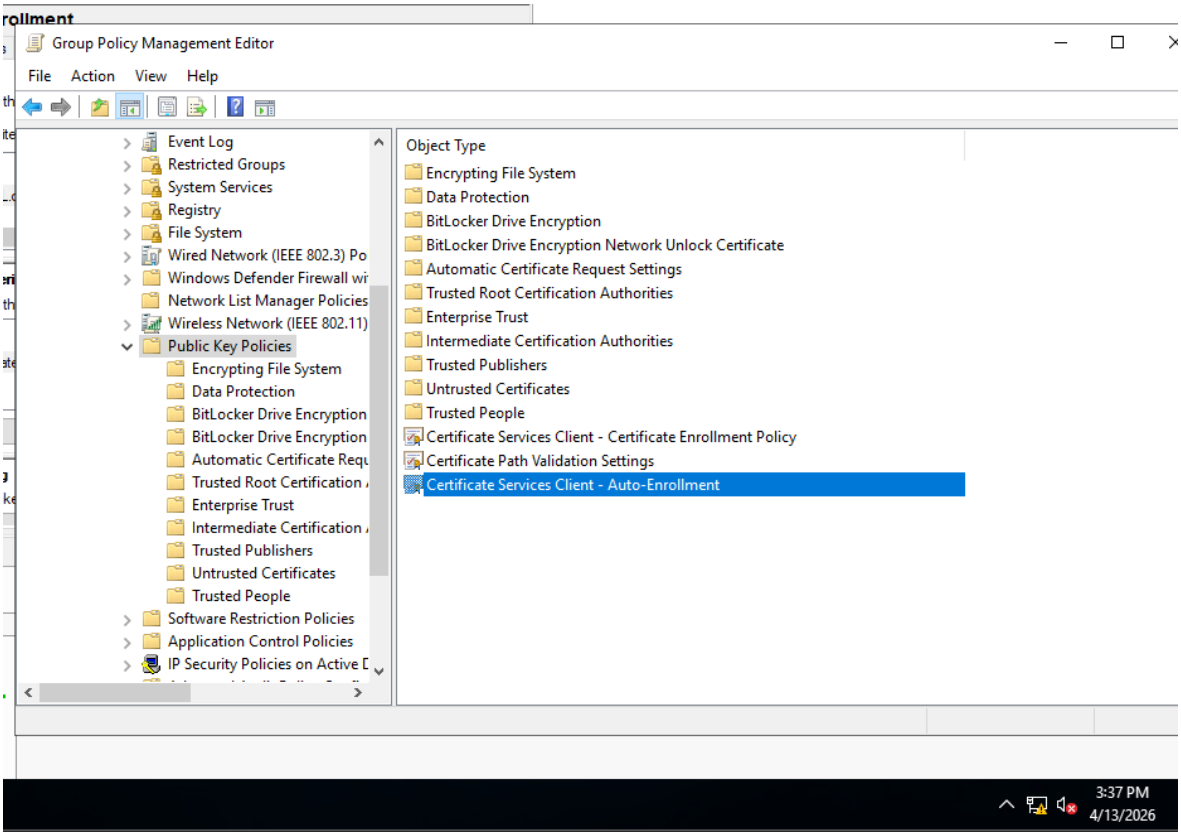




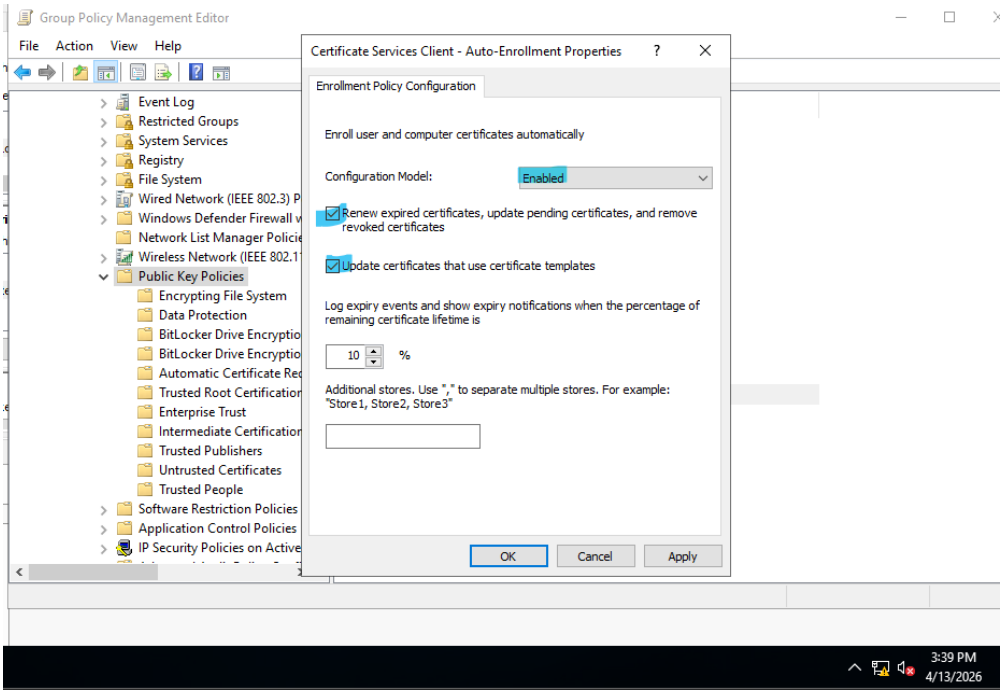
4. Créez une GPO : PKI-AutoEnrollment

- Link à Domain ou OU concernée.

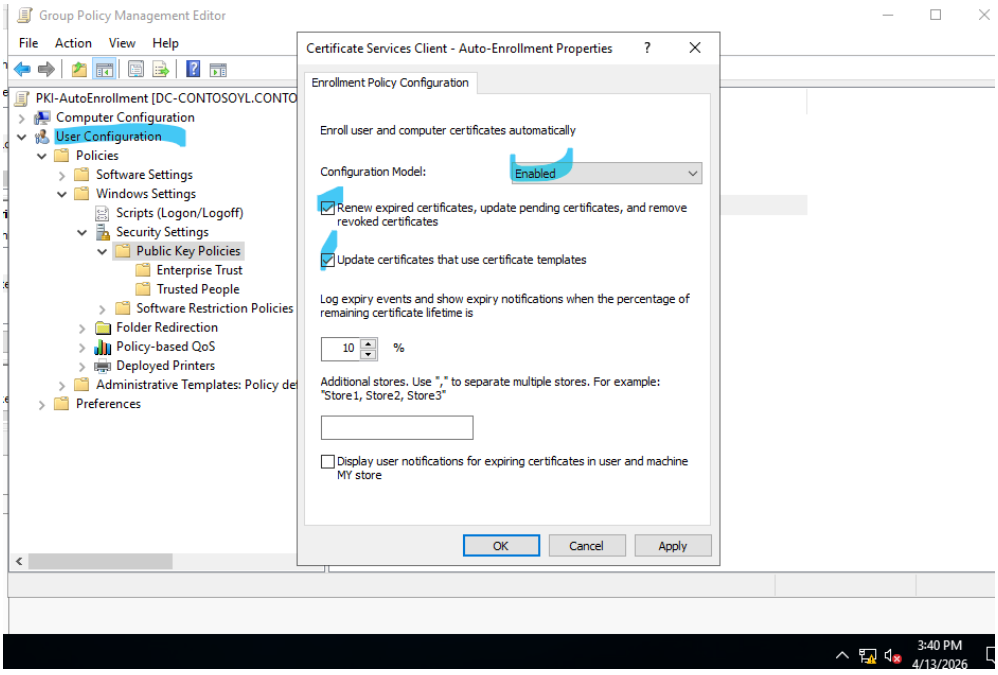
- Computer Configuration → Policies → Windows Settings → Security Settings → Public Key Policies :



- Certificate Services Client – Auto-Enrollment : Enabled + Renew expired + Update pending + Update using templates



- User Configuration : idem pour les utilisateurs.

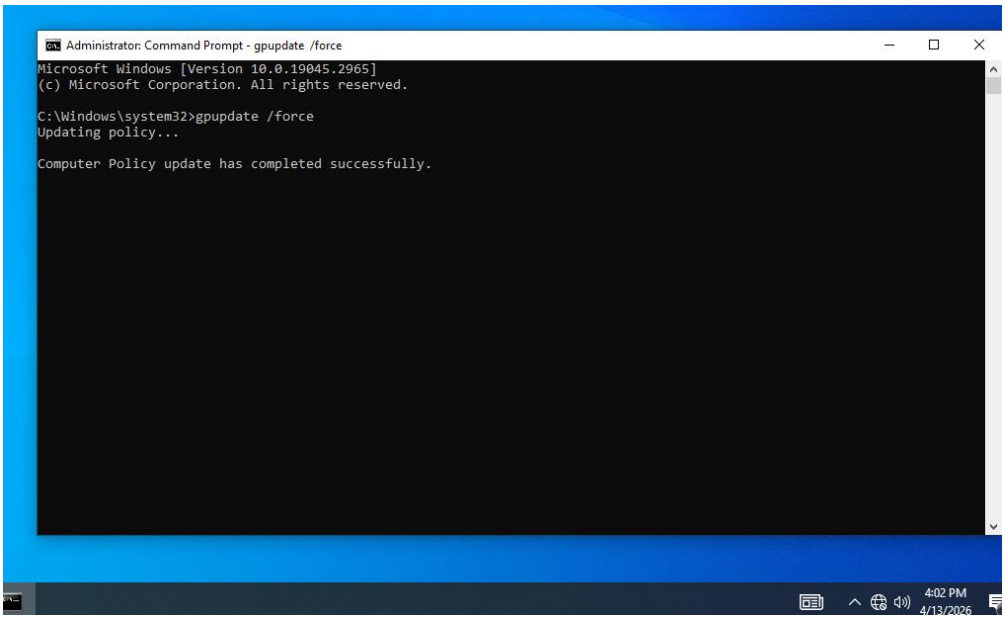


Section 9 – Tests d'enrollment depuis client Windows (10 pts)

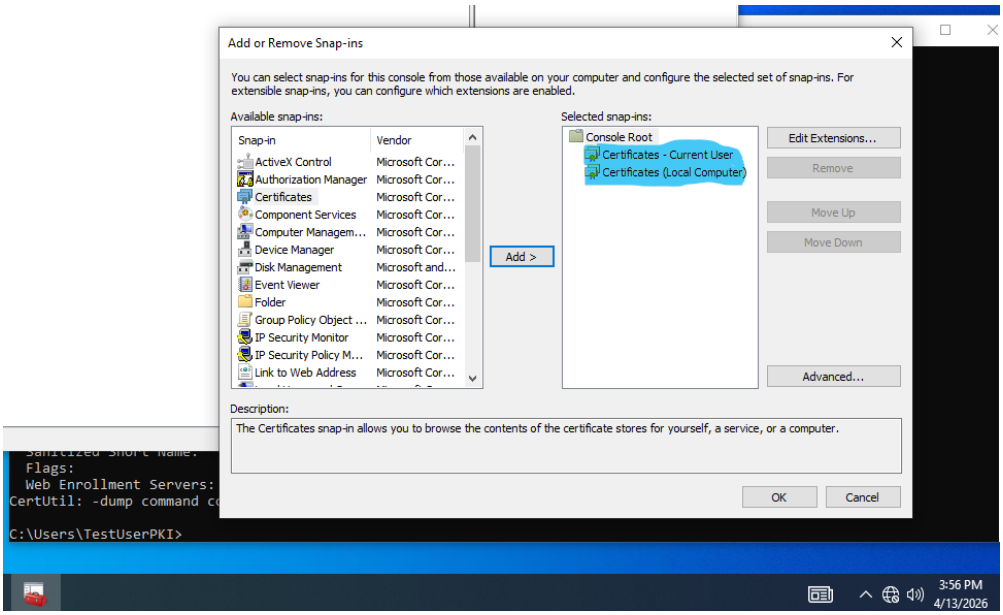
Sur le client Windows (TestUserPKI loggé) :

1. Auto-enrollment :

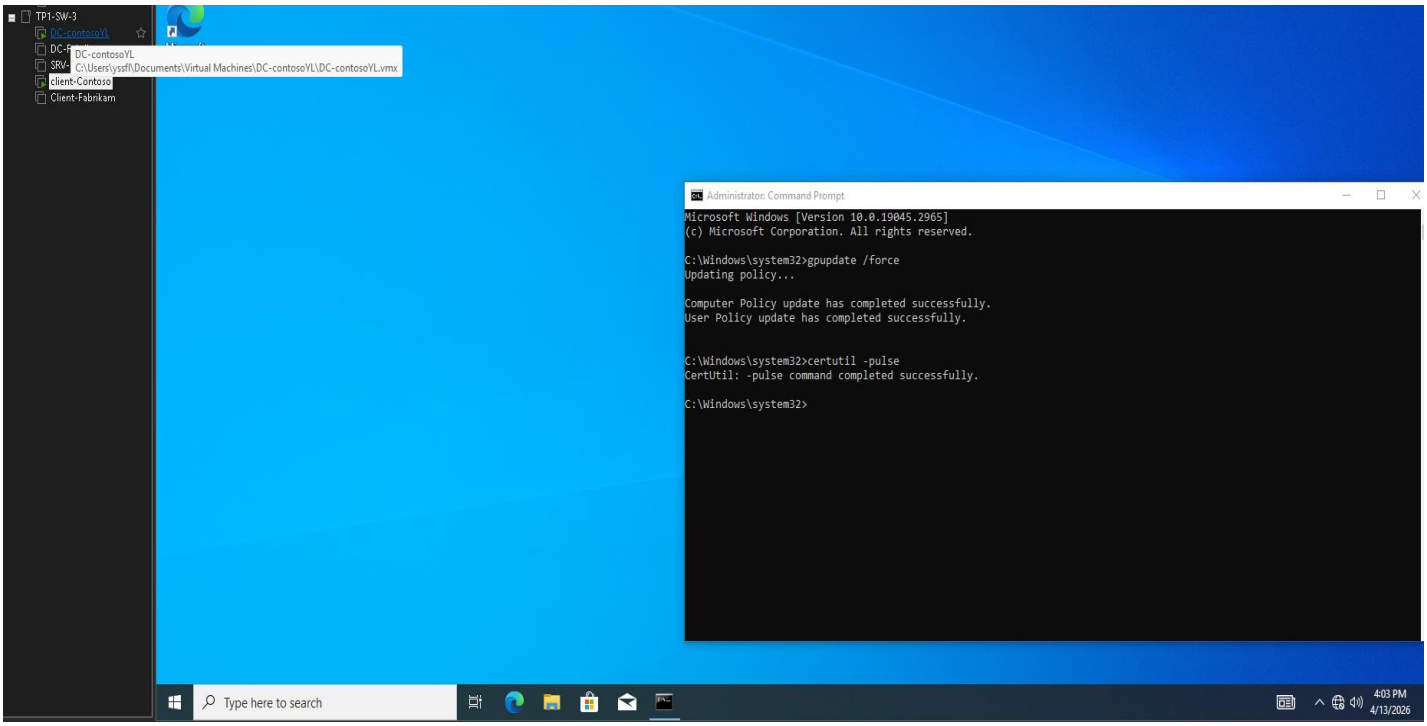
- Exécutez gpupdate /force



- Ouvrez mmc → Add Snap-in → Certificates - Current User / Computer

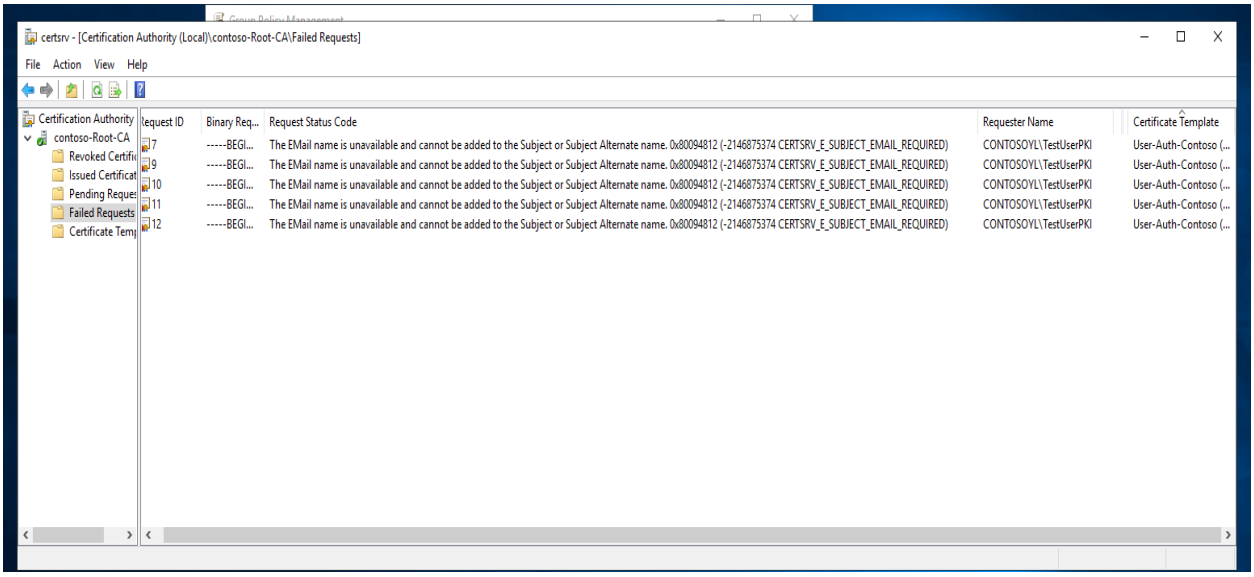


- Attendez 5–10 min ou forcez : certutil -pulse

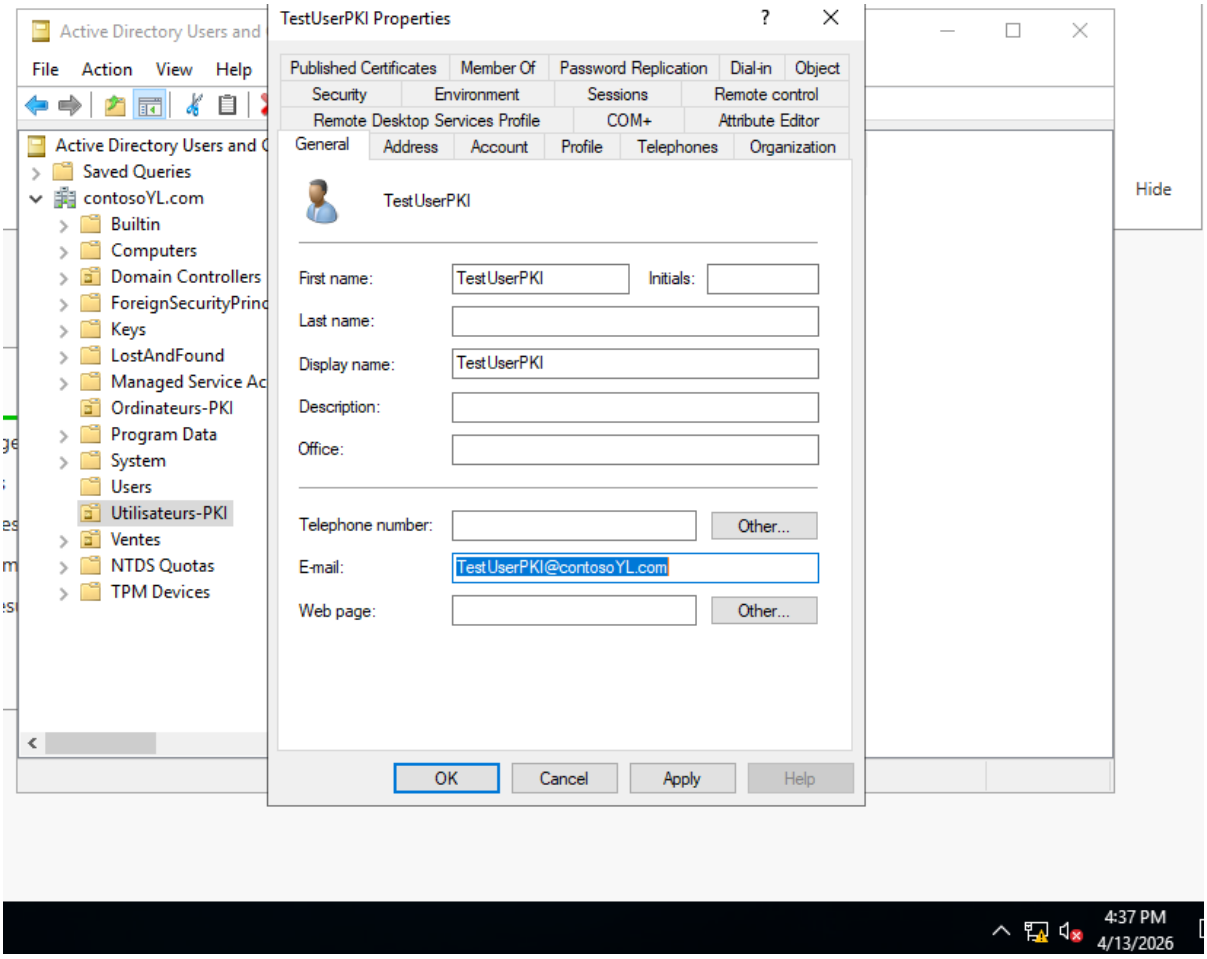


- Vérifiez si les certificats Computer-Auth-Contoso et User-Auth-Contoso apparaissent.

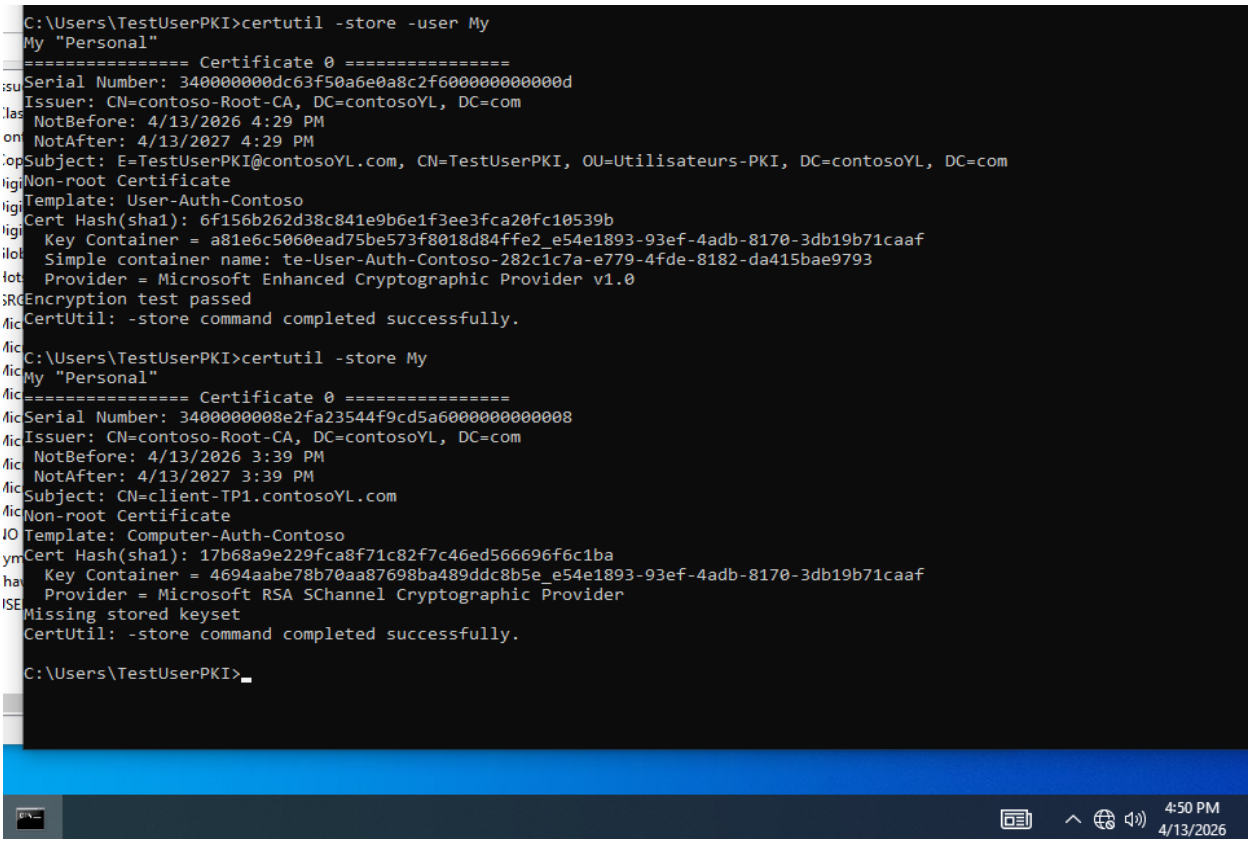
1. N’apparaisse pas mais j’ai remarqué une erreur dans CertAuth→Failed requests



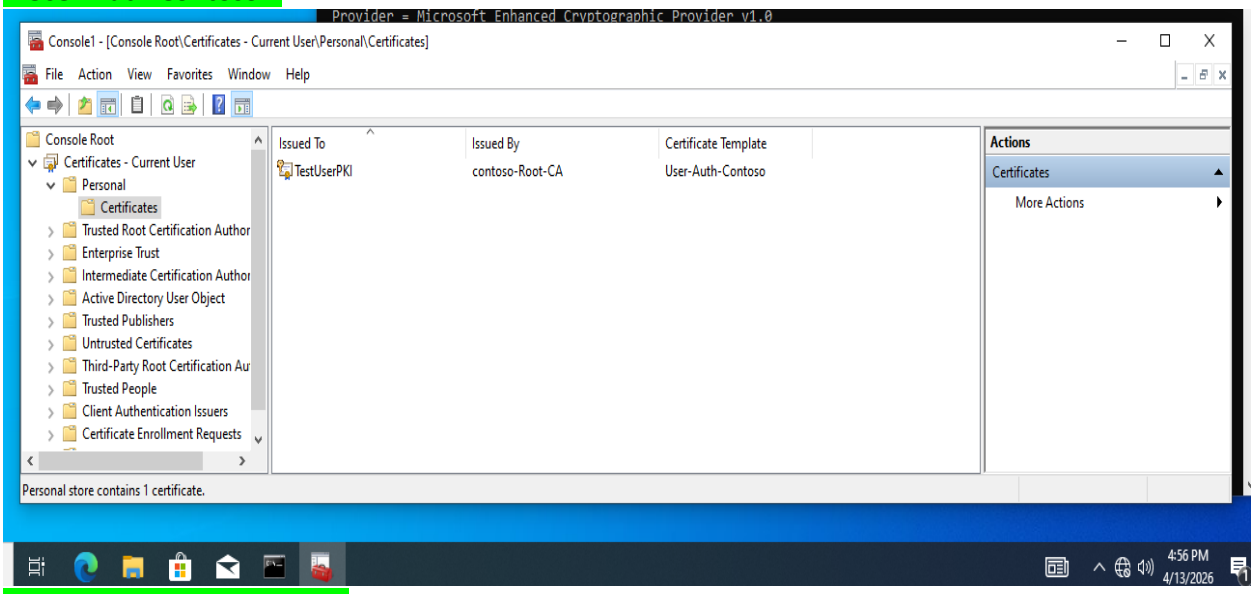
2. Deux options s'offraient à moi, soit donner un E-mail à mon **« TestUserPKI »** ou decocher la case « E-mail Name » dans Subject Name de **« User-Auth-Contoso »**
3. J'ai préféré aller avec la première option :
Dans le SRV-CS (contosoYL.com)→ ADUC→ j'ai ajouté un E-mail à l'utilisateur « TestUserPKI »



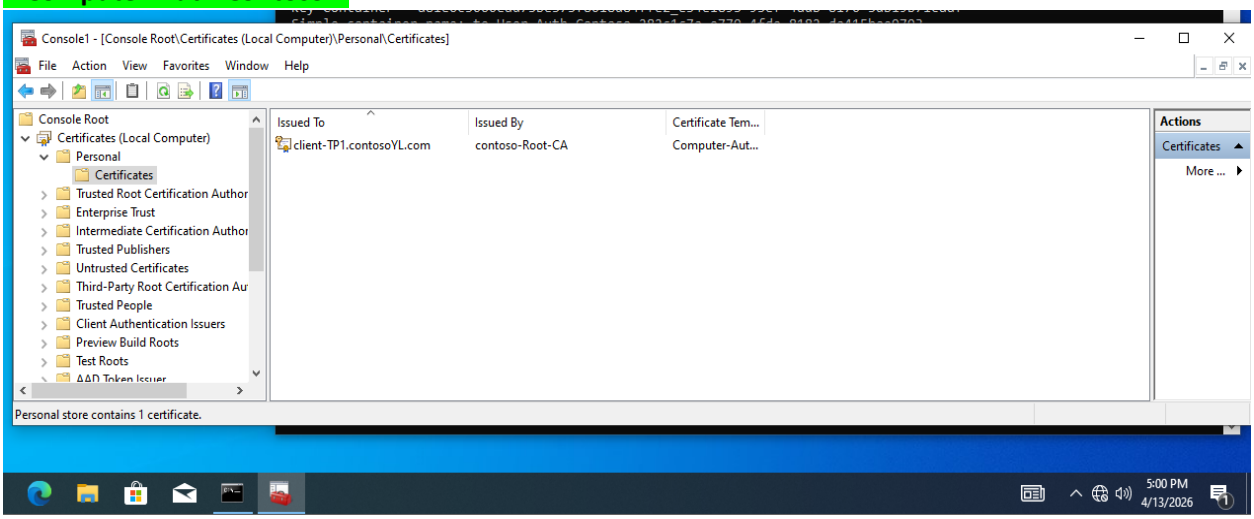
4. Ensuiter jai **« gpupdate /force » + « certutil -pulse » (en ADMIN) + « certutil -store My » et « certutil -store -user My (en USER)** (Cette commande liste les certificats de la machine et de l'utilisateur). Je devrais voir User-Auth-Contoso et Computer-Auth-Contoso apparaitre)



« User-Auth-Contoso »



« Computer-Auth-Contoso »



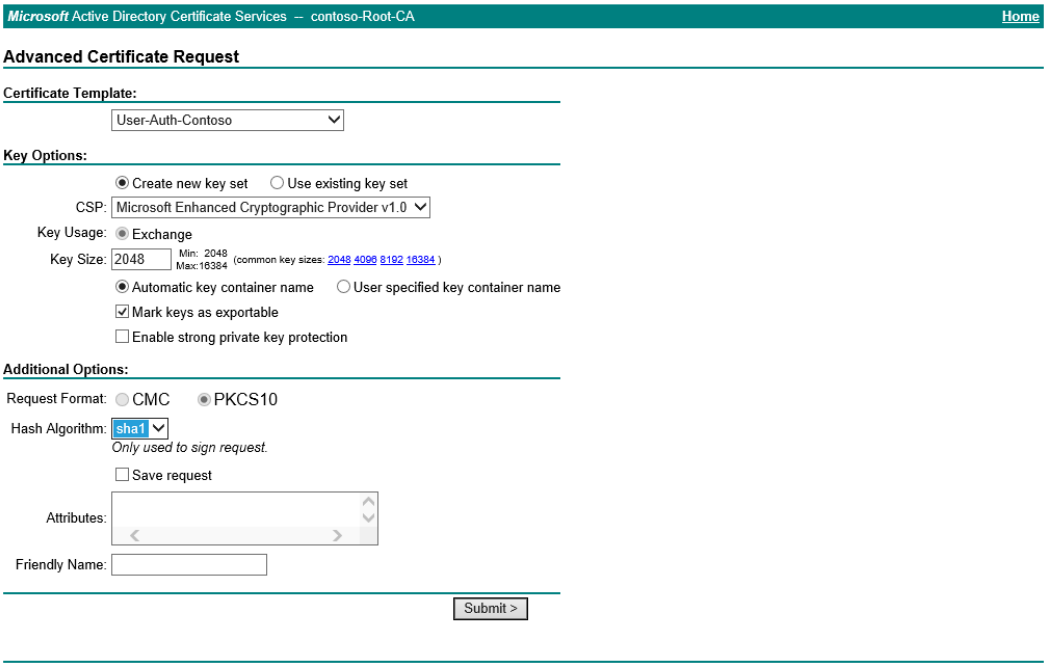
2. Enrollment manuel (via Web Enrollment) :

- Ouvrez IE/Edge : https://ca-server.contoso.com/certsrv (acceptez certificat si warning)

- Request a certificate → Advanced certificate request

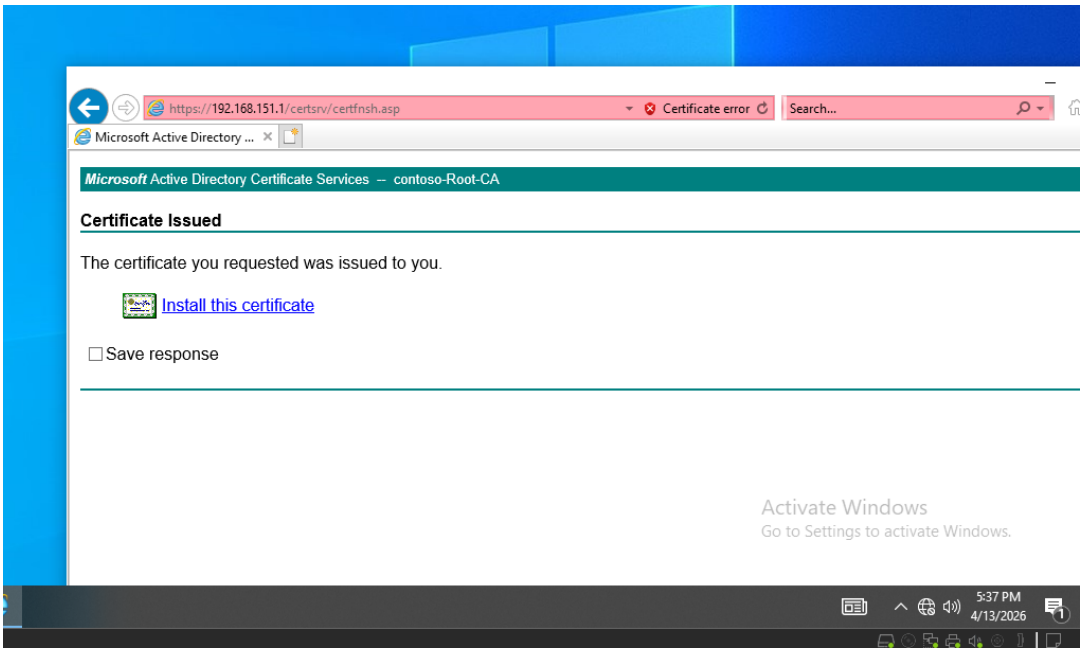
- Créez une demande pour template Computer ou User → Submit

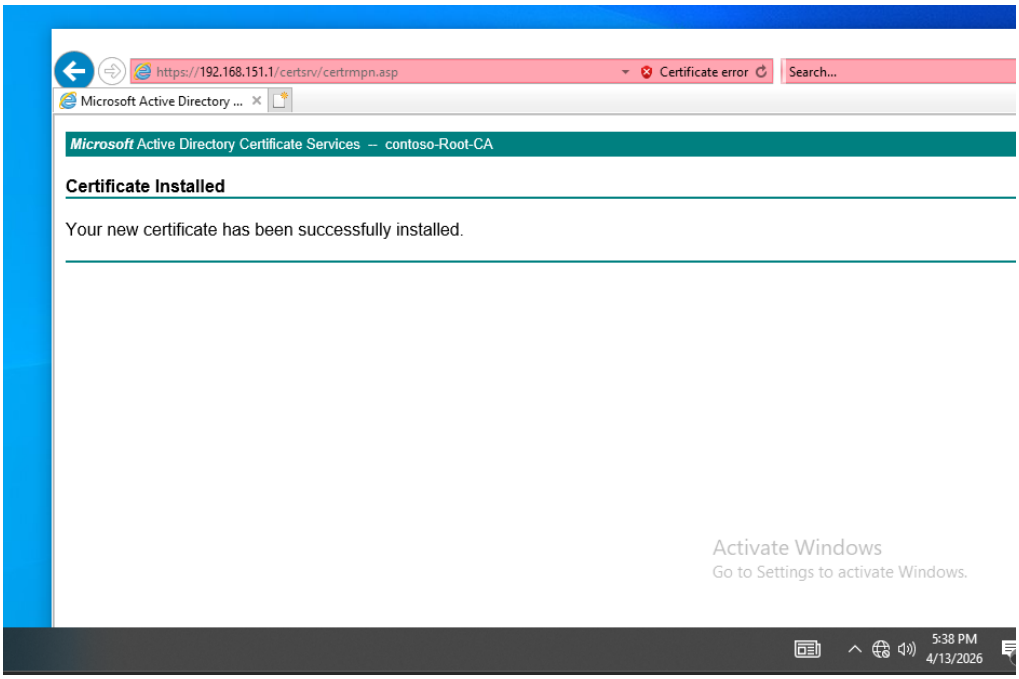
1. J'ai dû ajouter un binding (443-https) dans IIS (DC-contoso) pour pouvoir me connecter au **https://192.168.151.1/certsrv** sur le client





- Téléchargez/installez le certificat.

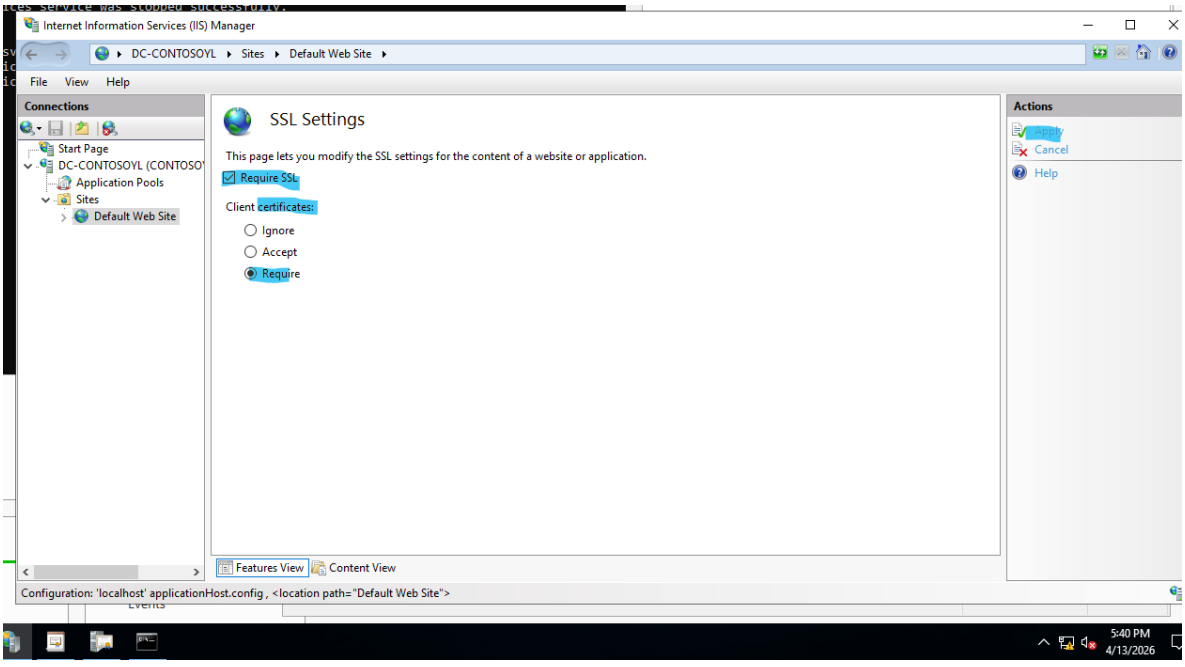




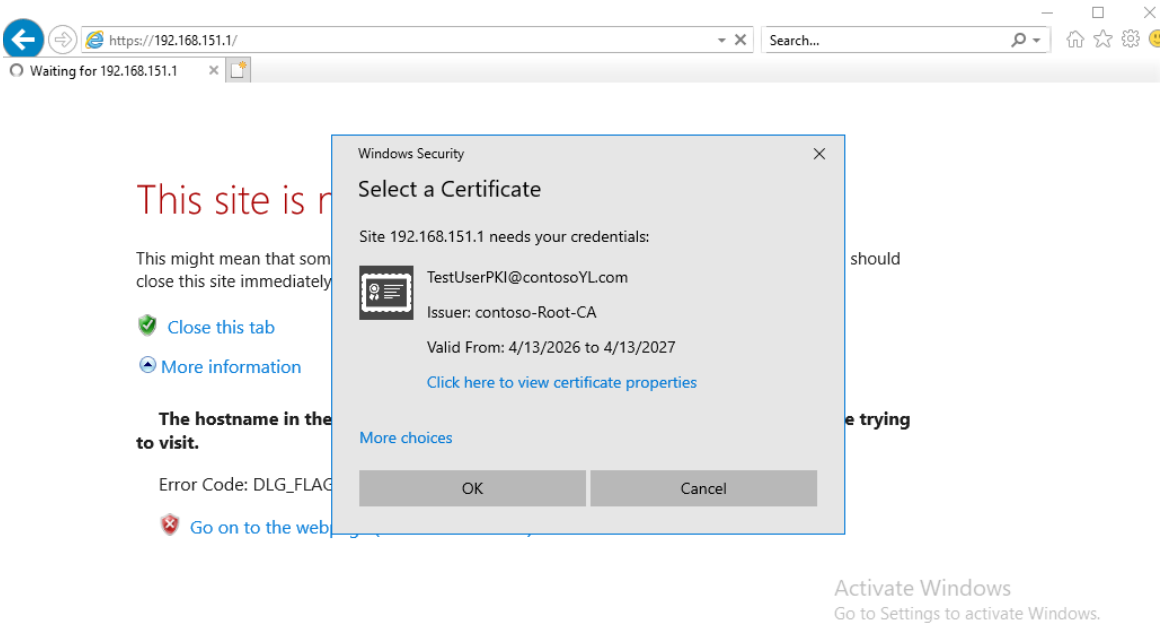
Section 10 – Tests d'utilisation et validation (10 pts)

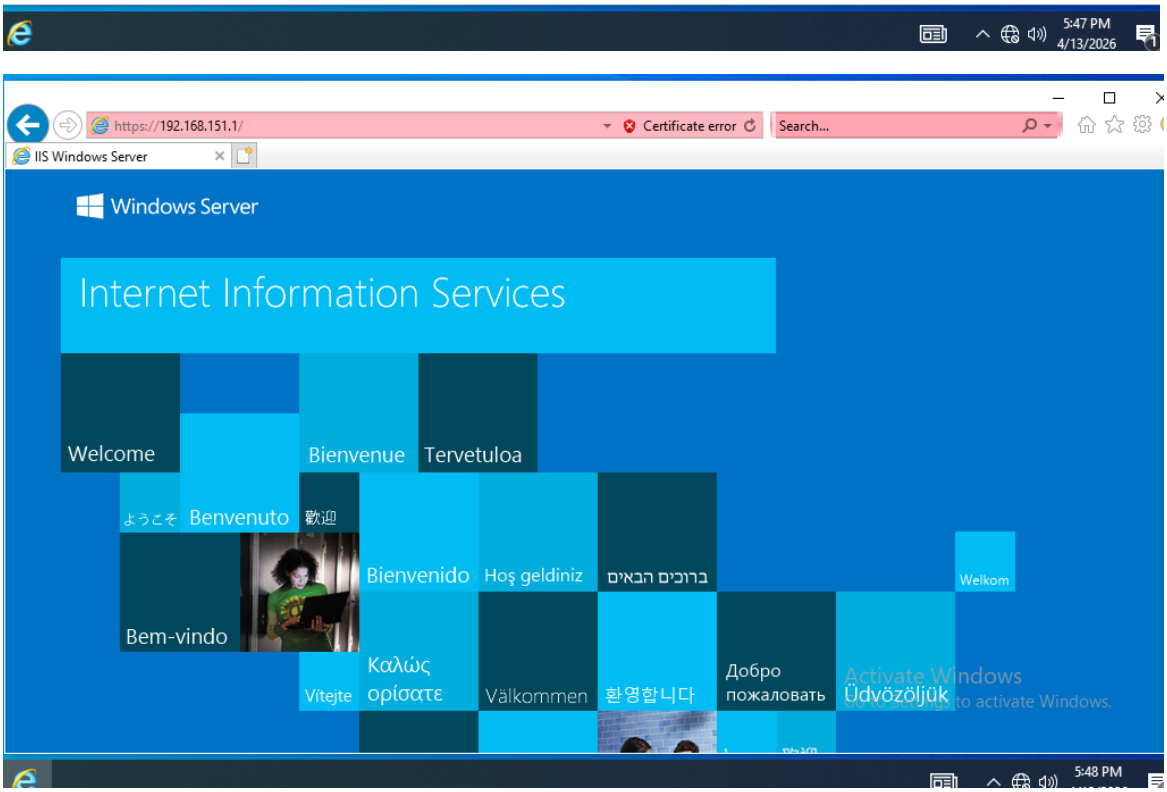
1. Test authentification client :

- Configurez un site IIS sur un serveur avec Require SSL + Client certificate.



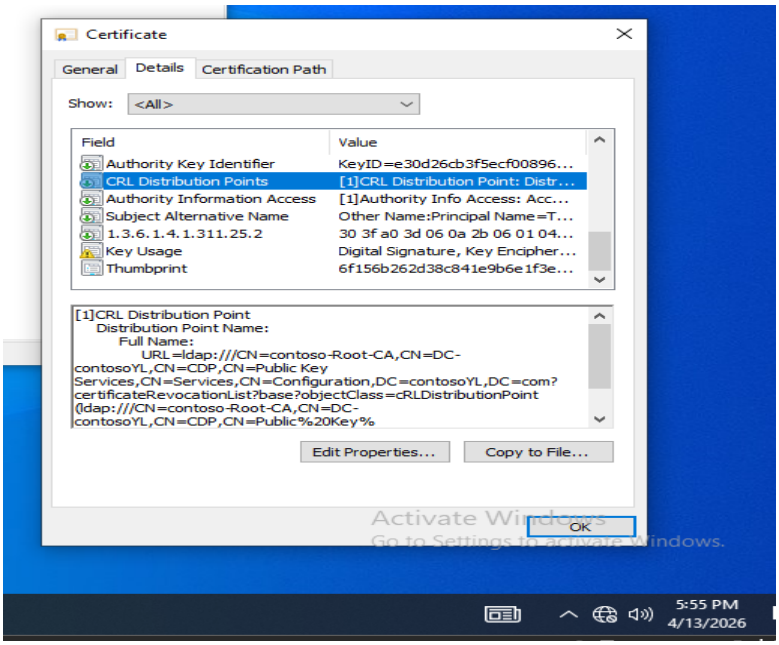
- Accédez depuis le client → sélectionnez certificat → accès OK.



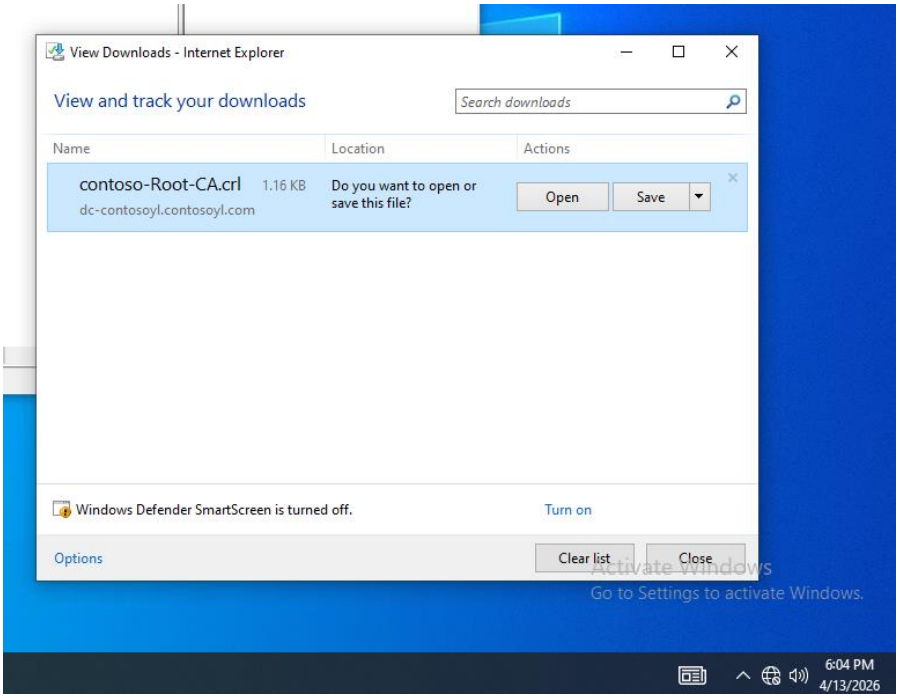


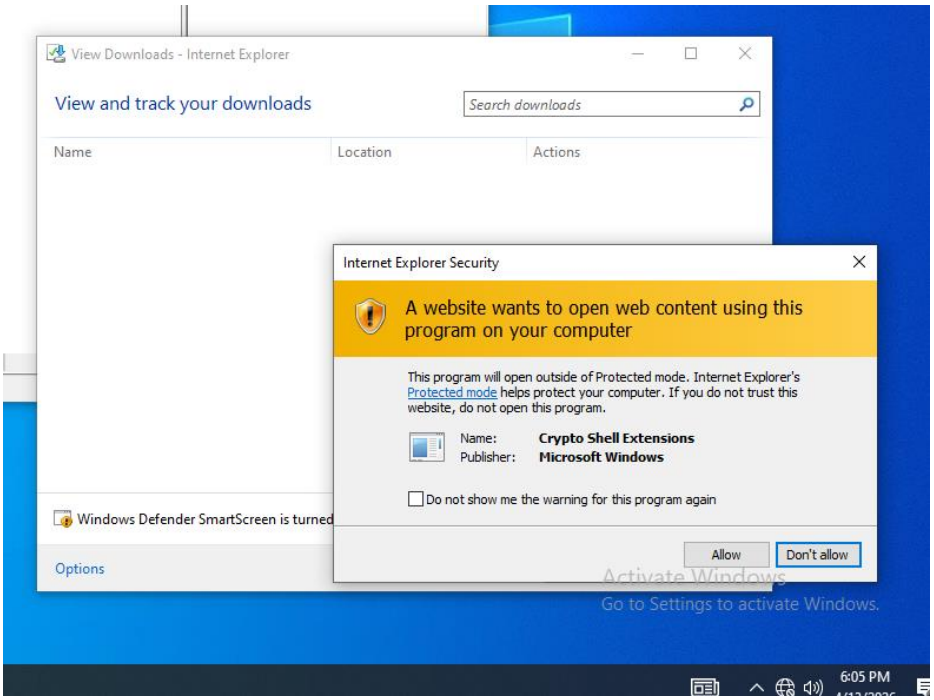
2. Vérifiez CRL/AIA :

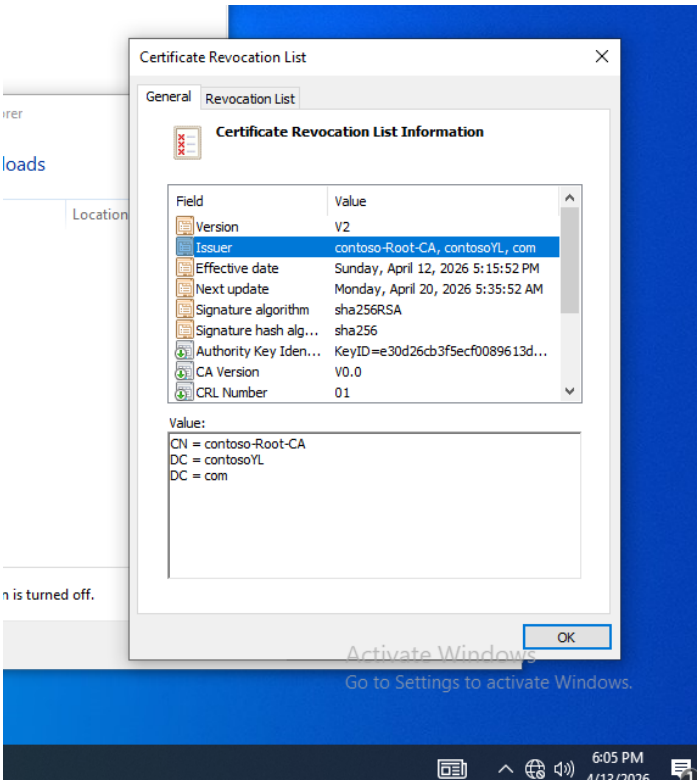
- Ouvrez un certificat → Certification Path → View Certificate → Details → CRL Distribution Points.



- Test : http://ca-server.contoso.com/crl (doit télécharger .crl).




View Downloads - Internet Explorer
View and track your downloads
Search downloads
Name
Location
Actions
Internet Explorer Security
A website wants to open web content using this program on your computer
This program will open outside of Protected mode. Internet Explorer's Protected mode helps protect your computer. If you do not trust this website, do not open this program.
Name: Crypto Shell Extensions
Publisher: Microsoft Windows
Do not show me the warning for this program again
Allow
Don't allow
Windows Defender SmartScreen is turned
Options
Activate Windows
Go to Settings to activate Windows.
6:05 PM


Certificate Revocation List
General
Revocation List
Certificate Revocation List Information
Field
Value
Version
V2
Issuer
contoso-Root-CA, contosoYL, com
Effective date
Sunday, April 12, 2026 5:15:52 PM
Next update
Monday, April 20, 2026 5:35:52 AM
Signature algorithm
sha256RSA
Signature hash alg...
sha256
Authority Key Iden...
KeyID=e30d26cb3f5ecf0089613d...
CA Version
V0.0
CRL Number
01
Value:
CN = contoso-Root-CA
DC = contosoYL
DC = com
OK
Activate Windows
Go to Settings to activate Windows.
6:05 PM